where

$$F_P(\theta) = \sqrt{\tan 2\theta / 2\theta}$$

$$F_{MA}(\theta) = \left(1 - 0.5\overline{\theta} - 0.05\overline{\theta}^2\right) \Big/ \sqrt{1 - \overline{\theta}}$$

$$F_{MB}(\theta) = \theta\left(0.5 - 0.25\overline{\theta} + 0.323\overline{\theta}^2\right) \Big/ \sqrt{1 - \overline{\theta}}$$

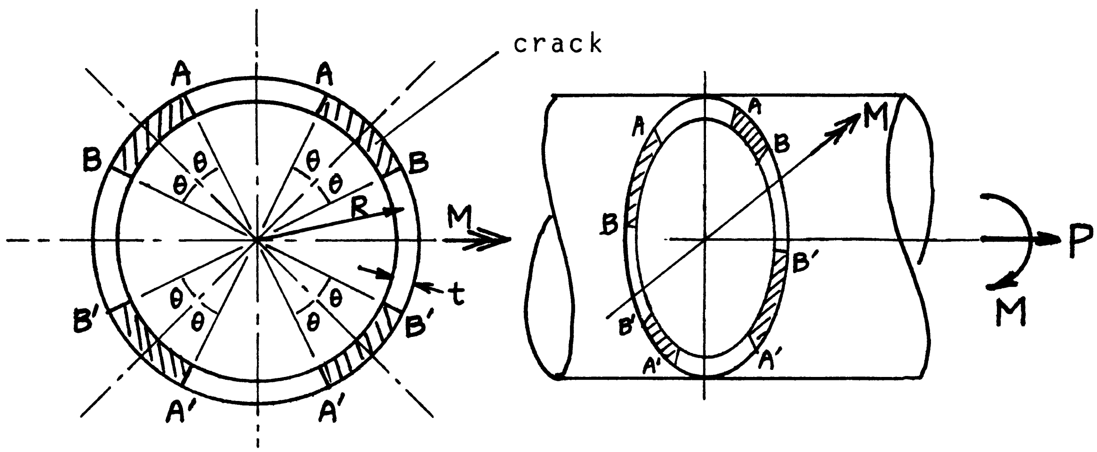
$$I_P(\theta) = \ell n(\sec 2\theta)$$

$$I_M(\theta) = \frac{1}{\sqrt{2}}\left\{\ell n \frac{1}{1 - \overline{\theta}} - \overline{\theta}\left(1 - 1.245\overline{\theta} + .333\overline{\theta}^2 - .044\overline{\theta}^3 - .005\overline{\theta}^4\right)\right\}$$

$$\overline{\theta} = \theta / \left(\frac{\pi}{4}\right) = \frac{4}{\pi}\theta$$

Method: K Approximated with Periodic Cracks (for P), Estimated by Interpolation (for M); A Paris' Equation (see **Appendix B**)
Accuracy: K Expected to be better than 5%; A Expected to be better than 10%
Reference: **Tada 2000**

NOTE: No crack surface interference in compressive region is assumed.



R = mean radius
t = wall thickness $\left(R/_t > 5\right)$

$$\sigma_P = P/(2\pi Rt) \qquad \sigma_M = M/\left(\pi R^2 t\right)$$

$$K_{I\binom{A}{A'}} = \sqrt{\pi R\theta}\left\{\sigma_P F_P(\theta)(\pm)\sigma_M F_{MA}(\theta)\right\}$$

$$K_{I\binom{B}{B'}} = \sqrt{\pi R\theta}\left\{\sigma_P F_P(\theta)(\pm)\sigma_M F_{MB}(\theta)\right\}$$

Crack Opening Area: $A = A_P + A_M$

$$A_{\binom{\widehat{AB}}{\widehat{A'B'}}} = A_P(\pm)A_{M\widehat{AB}}$$

$$= \frac{\pi R^2}{E'}\left\{\sigma_P I_P(\theta)(\pm)\sigma_M I_M(\theta)\right\}$$

where

$$F_P(\theta) = \sqrt{\tan 2\theta / 2\theta}$$

$$F_{MA}(\theta) = \left(0.707 - 0.070\overline{\theta}\right) \Big/ \sqrt{1-\overline{\theta}}$$

$$F_{MB}(\theta) = \left(0.707 + 0.076\overline{\theta} - 0.450\overline{\theta}^{\,2}\right) \cdot \sqrt{1-\overline{\theta}}$$

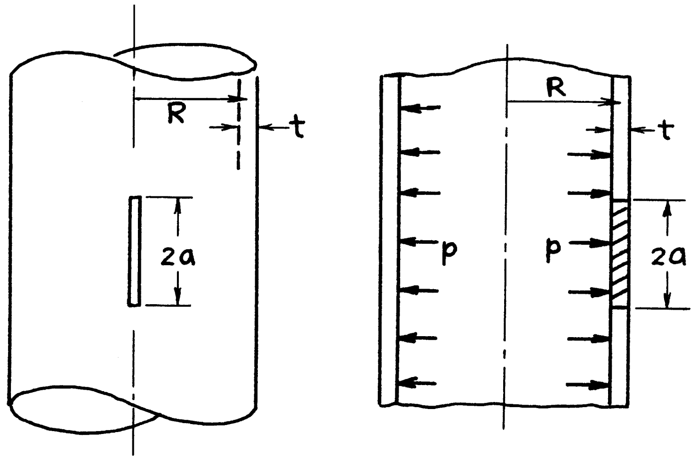
$$I_P(\theta) = \ell n(\sec 2\theta)$$

$$I_M(\theta) = \frac{1}{2}\left\{\ell n \frac{1}{1-\overline{\theta}} - \overline{\theta}\left(1 - 1.244\overline{\theta} + .329\overline{\theta}^{\,2} + .235\overline{\theta}^{\,3} - .116\overline{\theta}^{\,4} + .025\overline{\theta}^{\,5}\right)\right\}$$

$$\overline{\theta} = \theta / \left(\frac{\pi}{4}\right) = \frac{4}{\pi}\theta$$

Method: K Approximated with Periodic Cracks (for P), Estimated by Interpolation (for M); A Paris' Equation (see **Appendix B**)
Accuracy: K Expected to be better than 5%; A Expected to be better than 10%
Reference: **Tada 2000**

NOTE: No crack surface interference in compressive region is assumed.



$$\sigma = p\frac{R}{t}$$

$$\lambda = a\Big/\sqrt{Rt}$$

$$K_I = \sigma\sqrt{\pi a}\cdot F(\lambda)$$

$$F(\lambda) = (1 + 1.25\lambda^2)^{1/2} \qquad 0 < \lambda \leq 1$$

$$= 0.6 + 0.9\,\lambda \qquad 1 \leq \lambda \leq 5$$

Crack Opening Area:

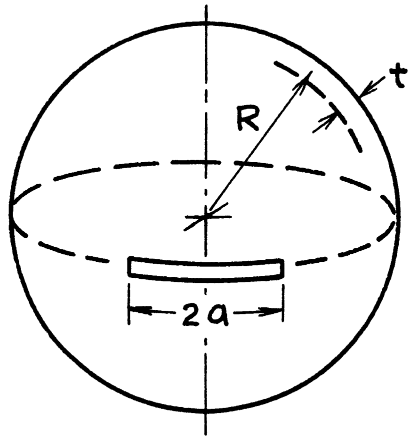
$$A = \frac{\sigma}{E'}(2\pi Rt)\cdot G(\lambda)$$

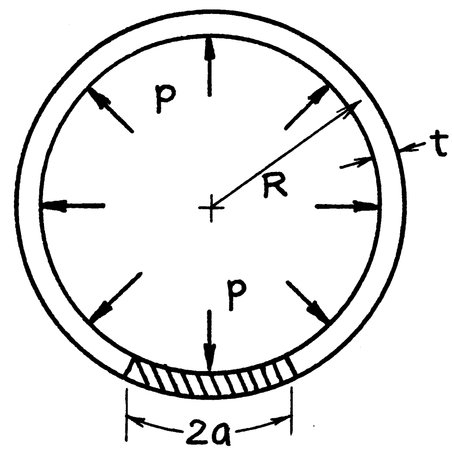
$$G(\lambda) = \lambda^2 + .625\lambda^4 \qquad 0 < \lambda \leq 1$$

$$= .14 + .36\lambda^2 + .72\lambda^3 + .405\lambda^4 \qquad 1 \leq \lambda \leq 5$$

Methods: K Integral Transform; A Paris' Equation (see **Appendix B**)
Accuracy: K_I 1%; A 2%
References: **Folias 1965; Erdogan 1969; Tada 1983a**

NOTE: As $a \to \infty$, $K_I \to \sigma\sqrt{R}\left(\frac{\sqrt{27\pi}}{2}\cdot\frac{R}{t}\right)$ (**Harris 1997**).





$$\sigma = \frac{1}{2} p \frac{R}{t}$$

$$\lambda = a \Big/ \sqrt{Rt}$$

$$K_I = \sigma\sqrt{\pi a} \cdot F(\lambda)$$

$$F(\lambda) = \left(1 + 1.41\lambda^2 + 0.04\lambda^3\right)^{1/2} \qquad 0 < \lambda \leq 3$$

Crack Opening Area:

$$A = \frac{\sigma}{E'}(2\pi Rt) \cdot G(\lambda)$$

$$G(\lambda) = \lambda^2 + 0.705\lambda^4 + 0.016\lambda^5 \qquad 0 < \lambda \leq 3$$

Methods: K_I Integral Equations; A Paris' Equation (see **Appendix B**)
Accuracy: K_I 1%; A 2%
References: **Erdogan 1969; Tada 1985**

APPENDIX A

COMPLIANCE CALIBRATION METHODS

A. DETERMINATIONS OF $\mathcal{G}$ AND K^2

As defined earlier and in **Irwin (1954)**, the crack extension force, $\mathcal{G}$, can be given by

$$\mathcal{G} = -\frac{dU_T}{dA} \text{(system isolated)} \quad \textbf{(A1)}$$

where U_T is the total strain energy of elastic deformation in the solid containing a crack, and dA is an infinitesimal (virtual) increment of new severed area. The system isolated requirement can be removed (**Paris 1965**) by subtracting from U_T the increment of energy, $\sum P_i d\Delta_i$, which enters the body during the movement, dA, of the crack. P_i is one of the applied loads and Δ_i is the corresponding load displacement (parallel to the load). Thus, $\mathcal{G}$ is given by

$$\mathcal{G} = \sum P_i \frac{d\Delta_i}{dA} - \frac{dU_T}{dA} \quad \textbf{(A2)}$$

Because the generalization from a single pair of loading points with oppositely directed loads, P, to multiple loading points is obvious, consider **Eq. (A2)** in the simpler form

$$\mathcal{G}\, dA = P d\Delta - dU_T \quad \textbf{(A3)}$$

Assuming linear-elastic behavior, the load point displacement, Δ (total between the pair of loading points) and the strain energy, U_T, are given by

$$\Delta = CP \quad \textbf{(A4)}$$

where C is the compliance, and

$$U_T = \frac{1}{2}\Delta P \quad \textbf{(A5)}$$

Substitution of (A4) and (A5) into (A3) provides

$$\mathcal{G}\,dA = \frac{1}{2}P^2\,dC \tag{A6}$$

Note that **Eq. (A5)** implies that the plate specimen does not contain a system of self-balanced (residual) stresses. In the case of a plate-type specimen of thickness B, containing only one straight leading edge of the crack, and taking the crack size to be a, dA equals Bda.
Thus **Eq. (A6)** becomes

$$\mathcal{G} = \frac{1}{2}P^2\,\frac{dC}{Bda} \tag{A7}$$

Further, if the plate-type specimen has a characteristic width W, and a Young's Modulus, E, **Eq. (A7)** can then be written in the form [see **Eq. (20)**]

$$K^2 = E\mathcal{G} = \frac{1}{2}\left(\frac{P}{BW}\right)^2 W\,\frac{d(EBC)}{d\left(^a/_w\right)} \tag{A8}$$

where the derivative term in the equation is dimensionless.

Eq. (A7) suggests an experimental method for determining the proportionality of $\mathcal{G}$ to P^2. Measurements of the ratio $\Delta/P = C$ for a series of crack lengths (a-values) can be made over a range of interest. The required derivative dC/da, as a function of a, can than be obtained from these results. Early trials of this method for notched round bars and edge notched bars in bending are found in **Lubahn (1959)**. In later efforts (**Srawley 1964**), a long, single-edge-notched plate in tension was used and the results were directly compared to numerically computed values. In terms of the information necessary for the computation of K values, it appeared that an accuracy of better than 3% could be achieved by the experimental method in central portions of the series of crack lengths used in the calibration. Potential improvement in accuracy of the compliance calibration method is to be expected with increases of dC/da relative to the value of C. A list of fracture test specimens, in order of decreasing expected potential compliance calibration accuracy, is:

1. Double-cantilever specimens
2. Single-edge-notched bend bars
3. Compact tension specimens
4. Long edge-notched plates in tension
5. Long centrally notched plates in tension

With regard to accuracy, since the end result required is generally a K calibration, and K^2 does not depend E, a low modulus plate material such as a high-strength aluminum alloy can be used to increase the observed displacements. However, the value of E for the calibration material must be adequately known or measured.* Comments on the accuracy of various displacement gages are given in **Srawley (1965)**. Contributions of plastic deformation at the loading points should be eliminated and influences of any plastic strain at the tip of the crack-simulating notch should be minimized. After establishing calibration, values of K can be obtained from various pairs of observations such as P and a, P and C, and Δ and a.

* **It is not sufficient to use a literature value.**

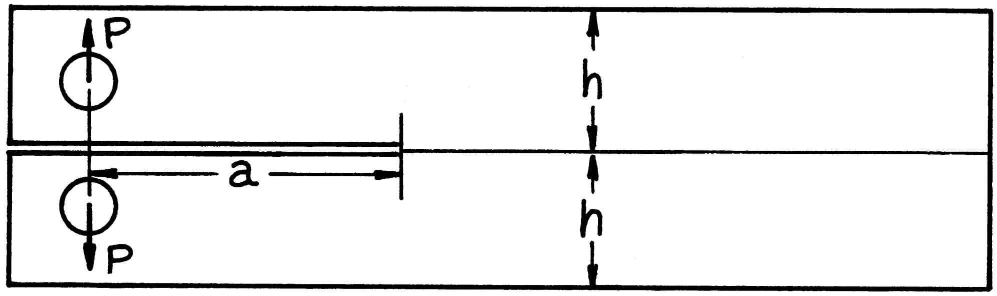


Fig. (A1). DCB specimen for measuring the tensile fracture toughness of an adhesive bond.

Eq. (A7) and (A8), coupled with compliance measurements, have been frequently used for double-cantilever (DCB) specimens. The conditions pertaining to the use of DCB specimens for measurements of tensile fracture strength of adhesive joints and, with face-grooving, for homogeneous solids are reviewed in **Irwin (1971)**. Consider two adherent bars joined by a thin layer of adhesive as shown in **Fig. (A1)**.

If an estimate of C is made using simple beam theory with a correction for deflection due to shear, the following expression for $\mathcal{G}$ can be derived.

$$\mathcal{G} = \frac{4P^2}{EhB^2}\left\{3\left(\frac{a+a_0}{h}\right)^2 + 1\right\} \tag{A9}$$

Eq. (A9) assumes each beam is "built-in" at a distance $(a + a_0)$ from the loading line. The second term in the bracketed expression results from the shear correction and takes the simple form shown if one assumes Poisson's ratio is 1/3. Experimental results suggest a value of about *h/3* for a_0. These assumptions appear to be adequate for estimates of $\mathcal{G}$ so long as a is larger than *2h* and the unbroken ligament is larger than *3h*. However, a calibration of improved accuracy can be readily developed through careful compliance measurements (**Ripling 1964**).

Ripling (1964) discusses a modified (tapered) DCB specimen shape such that dC/da does not depend crack size across a certain range of a values. In the case of all of the DCB specimens discussed above, the experimentors have relied primarily compliance calibrations rather than results derived from numerical analysis of the stress fields.

Compliance calibration methods are applicable to plate-type specimens with through-the-thickness cracks for which thickness average values of $\mathcal{G}$ are sufficient for experimental purposes. Compliance calibration can be used, with limited accuracy, for notched round bars in tension because $\mathcal{G}$ is constant around the leading edge of the crack-simulating notch. In the case of a crack for which the variation of $\mathcal{G}$ along various regions of the leading edge is appreciable and important, it must be remembered that measurements of the change of compliance with enlargement of the crack can provide only $\mathcal{G}$ value averages. With regard to the relationship between the thickness average $\mathcal{G}$ and K^2, the assumption, $K^2 = E\,\mathcal{G}$, implied in the K_{IC} test method, ASTM E-399, can be regarded as adequate for practical applications.

When a plate specimen is used as a model of a service component that contains a system of self-balanced (residual) stresses, attention must be given to the possibility that the residual stresses may hold the crack partially closed during initial portions of the externally applied loading. In the absence of this complexity, a plate specimen can be used to determine that portion of K that is due to the external loading. The remaining portion of K, due to the residual stresses, can be found from knowledge of the residual stresses using the analytical or numerical methods discussed in **Part 1**.

B. RELATIONSHIP OF $\mathcal{G}$ TO J

In this section, J is given a restricted interpretation as merely an expression for $\mathcal{G}$ applicable to a non-linear-elastic as well as to a linear-elastic solid containing a crack. Thus **Eqs. (A1) and (A2)** may be regarded as definitions of J (**Rice 1968a**). For simplicity this discussion is restricted to two-dimensional crack stress fields. **Eq. (A2)** suggests a plan for representing J as a path-independent contour integral enclosing the crack tip, as follows. Because the solid is elastic, energy disappearance can occur only at the crack tip singularity during its forward increment of (virtual) displacement. Thus the net flux of energy inward across the boundary of any area enclosing the crack tip must give the same value, J. In the construction of a J contour integral, the summation of input work, $\sum P_i d\Delta_i$, can be replaced by its equivalent in terms of the integral of each boundary stress times the parallel increment of displacement for the individual segments of the contour. In addition the change of stress field energy that occurs inside of the contour must be subtracted.

Assuming an increment, da, of forward (x-direction) displacement of the crack tip, and assuming the coordinate normal to the crack is y, one finds

$$Jda = -da\iint dxdy\left(\frac{\partial U}{\partial a}\right) + da\oint\left[dy\left(\sigma_x\frac{\partial u}{\partial a} + \tau_{xy}\frac{\partial v}{\partial a}\right) - dx\left(\tau_{xy}\frac{\partial u}{\partial a} + \sigma_y\frac{\partial v}{\partial a}\right)\right] \qquad \textbf{(A10)}$$

where U is the strain energy density.

Obviously da can be eliminated from each side of **Eq. (A10)**. Next, consider the fact that the location of the contour relative to the crack tip, after the crack tip has been displaced an infinitesimal amount da in the x-direction, is equivalent to rigid displacement of the contour by an equal infinitesimal amount, dx, in the opposite direction. Thus the calculations indicated in **Eq. (A10)** are unchanged if $\frac{\partial}{\partial a}$ is replaced by $-\frac{\partial}{\partial x}$. With regard to the preceding term of **Eq. (A10)**, note that the substitution of $-\frac{\partial}{\partial x}$ for $\frac{\partial}{\partial a}$ permits completion of the x-direction portion of the integral or

$$\iint dxdy\left(\frac{\partial U}{\partial x}\right) = \oint U\,dy \qquad \textbf{(A11)}$$

The final result is

$$J = \oint\left[dy\left\{U - \sigma_x\frac{\partial u}{\partial x} - \tau_{xy}\frac{\partial v}{\partial x}\right\} + dx\left\{\tau_{xy}\frac{\partial u}{\partial x} + \sigma_y\frac{\partial v}{\partial x}\right\}\right] \qquad \textbf{(A12)}$$

Eq. (A12) can be expressed in more compact form. However, the form given above is convenient for purposes of numerical computation. As a check on the path independence of **Eq. (A12)**, one can equate $\frac{\partial}{\partial x}$ of the coefficient of dy to $\frac{\partial}{\partial y}$ of the coefficient of dx. After use of stress compatibility relations, the result is

$$\frac{\partial U}{\partial x} = \sigma_x\frac{\partial \varepsilon_x}{\partial x} + \sigma_y\frac{\partial \varepsilon_y}{\partial x} + \tau_{xy}\frac{\partial \gamma_{xy}}{\partial x} \qquad \textbf{(A13)}$$

Given the existence of a set of stress–strain relations, linear or nonlinear, which are strain-path independent, U can be represented as

$$U = \int_0^{\varepsilon_x}\sigma_x d\varepsilon_x' + \int_0^{\varepsilon_y}\sigma_y d\varepsilon_y' + \int_0^{\gamma_{xy}}\tau_{xy} d\gamma_{xy}' \qquad \textbf{(A14)}$$

From the assumptions, the integrand of each integral can be regarded as a function only of the corresponding integration variable, ε_x', ε_y' or γ_{xy}'. The derivative of **Eq. (A14)** then results in **Eq. (A13)**.

In later applications, the path independence of **Eq. (A12)** will be of special interest because the path of integration can be taken around the fracture process zone, well within the region of pronounced nonlinearity, yet outside the region containing advance separations and localized slip displacements special to the progressive fracturing process. Thus J can serve as a characterization of the stress–strain environment enclosing the fracture process zone.

Anticipating a later discussion of J with plasticity present, without discussing applicability of J inside a plastic zone, it is clear that the J-Integral is path independent for any contour on or outside the elastic–plastic boundary. When the plastic zone of an opening mode crack is modeled as a Dugdale or strip-yield zone extending directly forward from the crack tip, it is easily shown that $J = \sigma_Y \delta$, where δ is nominally the crack opening stretch and σ_Y is a flow stress property assumed to be constant along the strip plastic zone. **Tada (1972a)** presents calculations that tend to support the likelihood of close equivalence between J and values of $\mathcal{G}$ computed using linear-elasticity with an adjustment of the crack size. Given a crack-tip plastic zone which does not contact a free surface of the specimen, other than the crack surfaces, and drawing the J-Integral contour as a circle enclosing the crack-tip plastic zone, calculation of $\mathcal{G}$ using linear-elastic assumptions but with the crack-tip moved into a central position within the plastic zone may give a result similar to the J-value.

Returning to compliance calibration methods, it is evident that **Eq. (A3)**, rather than its equivalent, in the form of **Eq. (A12)**, is of most interest as a basis for compliance calibration. Assuming only two oppositely directed loads, P, as before, a series of P versus Δ lines for a series of crack sizes, a, will not be straight lines, as with linear behavior. However, values of J times da (where da is the difference of the crack lengths for adjacent curves) can be determined as a function of Δ by careful planimeter measurements, by curve fitting and integration, and so on. In the case of nonlinear behaviors such that slope $dP/d\Delta$ is small in the region of measurement, it is most convenient to plot J as a function of Δ for the series of crack lengths of interest. The J determinations can then, if desired, be based observations of a and Δ.

C. CRACK SIZE DETERMINATIONS

Part A of this section discusses determinations of C as a function of crack size. In the case of DCB specimens, the sensitivity of Δ at fixed load to crack size is adequate for the purpose of using the Δ/P ratio to determine the crack size. In other specimens, determinations of crack size from displacement–load ratios are usually more accurate when a different choice is made of the displacement measurement position. For example, with a centrally notched tensile specimen, the opening at the notch center is a convenient and appropriate choice. In the case of long single-notched tensile specimens, the opening at the free edge intersection of the crack is often used. In order to use a displacement–load ratio to estimate crack size, an accurate calibration based measurements of this ratio in the range of linear behavior for a series of crack sizes can be used.

Moreover, many analytical expressions for these displacements are found in this handbook. In the case of thin-sheet testing to determine cracking resistance, restraints against out-of-plane deformation (buckling) may be advisable. Even when such restraints are used, the displacement-measuring device should be designed so as to indicate only center-plane displacement or so as to compensate for small amounts of out-of-plane bending.

Testing or application situations where the plastic zone size is appreciable relative to the crack size are not uncommon. In such cases it is a common practice to compute K on the basis of an effective crack size. The addition of r_Y to the crack size at each crack-tip involves uncertanities with regard to choice of α and σ_Y in the equation

$$r_Y = \alpha \left(\frac{K}{\sigma_Y} \right)^2 \qquad \textbf{(A15)}$$

As an alternative, one can determine the effective crack size from the "effective compliance," where the effective compliance is simply the ratio of deflection to load at the load for which a calculation of K is desired. Trials with analytical elastic–plastic models have shown that different choices of the displacement measurement position will cause differences in the effective crack size estimate. However, these differences are relatively small. Any disadvantage of this kind appears to be outweighed by the advantage of avoiding an arbitrary choice of α and σ_Y in the r_Y equation.

APPENDIX B

A METHOD FOR COMPUTING CERTAIN DISPLACEMENTS RELEVANT TO CRACK PROBLEMS

In earlier sections, $\mathcal{G}$ was shown (see **page 1.6**) to be equivalent to the rate of increase of the total strain energy, U_T, with increase in crack area, dA, and loading forces constant. That is to say

$$\mathcal{G} = \left.\frac{\partial U_T}{\partial A}\right|_{\text{Forces Constant}} \tag{B1}$$

Consider a body loaded by forces, P, which in addition has virtual forces, F, applied as in **Fig. (B1)**. (No generality is lost in considering a single force, P, and its support reaction and a single pair of forces, F, in equilibrium any distance, d, apart in the undeformed position.) The total energy rate, $\mathcal{G}$, can be, summed for the three modes, that is,

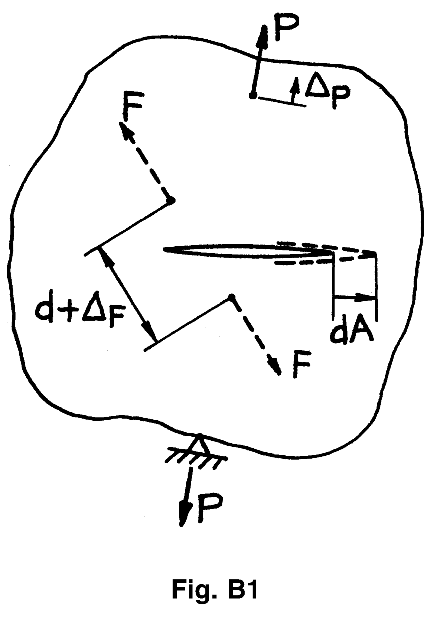


Fig. B1

$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II} + \mathcal{G}_{III} \tag{B2}$$

as was illustrated earlier (see **page 1.7**). Moreover, the resulting K_i values for each mode due to loading forces and virtual forces may be added, or

$$\left.\begin{aligned} K_I &= K_{IP} + K_{IF} \\ K_{II} &= K_{IIP} + K_{IIF} \\ K_{III} &= K_{IIIP} + K_{IIIF} \end{aligned}\right\} \tag{B3}$$

using the relationships between K_i and energy rates $\mathcal{G}_i$ (see **page 1.7**), that is,

$$E'\mathcal{G}_I = K_I^2, \quad E'\mathcal{G}_{II} = K_{II}^2, \quad E'\mathcal{G}_{III} = \alpha K_{III}^2 \tag{B4}$$

where E' and α are elastic constants depending on stress conditions, plane stress or plane strain. Combining **Eqs. (B2), (B3), and (B4)** gives

$$E'\mathcal{G} = (K_{IP} + K_{IF})^2 + (K_{IIP} + K_{IIF})^2 + (K_{IIIP} + K_{IIIF})^2 \cdot \alpha \tag{B5}$$

With the above information and Castigliano's theorem, it is possible to compute certain displacements in the manner suggested by **Paris (1957)**. Castigliano's theorem states that displacement of any load Q (in its own direction) may be computed by

$$\Delta_Q = \frac{\partial U_T}{\partial Q} \tag{B6}$$

The total strain energy may be regarded as that due to applying loading forces with no crack present plus that due to introducing the crack while holding forces constant, or

$$U_T = U_{No\,Crack} + \int_0^A \frac{\partial U_T}{\partial A} dA \tag{B7}$$

Introducing **Eqs. (B1) and (B6)**, displacement can be computed from

$$\Delta_Q = \frac{\partial U_T}{\partial Q} = \frac{\partial U_{No\,Crack}}{\partial Q} + \frac{\partial}{\partial Q}\int_0^A \mathcal{G} dA \tag{B8}$$

where it may be interpreted that

$$\frac{\partial U_{No\,Crack}}{\partial Q} = \Delta_{Q\,No\,Crack} \tag{B9}$$

Thus substituting **Eq. (B5) into (B8)**, and allowing the virtual forces to approach zero $(F \to 0)$, gives

$$\Delta_P = \Delta_{P\,No\,Crack} + \frac{2}{E'}\int_0^A \left(K_{IP}\frac{\partial K_{IP}}{\partial P} + K_{IIP}\frac{\partial K_{IIP}}{\partial P} + \alpha K_{IIIP}\frac{\partial K_{IIIP}}{\partial P}\right) dA$$

or (B10)

$$\Delta_F = \Delta_{F\,No\,Crack} + \frac{2}{E'}\int_0^A \left(K_{IP}\frac{\partial K_{IF}}{\partial F} + K_{IIP}\frac{\partial K_{IIF}}{\partial F} + \alpha K_{IIIP}\frac{\partial K_{IIIF}}{\partial F}\right) dA$$

Therefore displacements Δ_P or Δ_F may be computed from displacements with no crack present and a knowledge of K-formulas by using **Eq. (B10)**. This result adds interest in concentrated force K-formula solutions for forces such as F, as their displacement may be computed directly.

Special Case of Opening Displacements of Crack Surfaces

When computing relative displacements of crack surfaces, the displacements with no crack present are zero, that is,

$$\Delta_{No\,Crack} = 0 \tag{B11}$$

Moreover, if opening mode (Mode I) displacements are desired, as indicated in **Fig. (B2)**, then normally

$$\frac{\partial K_{IIF}}{\partial F} = \frac{\partial K_{IIIF}}{\partial F} = 0 \tag{B12}$$

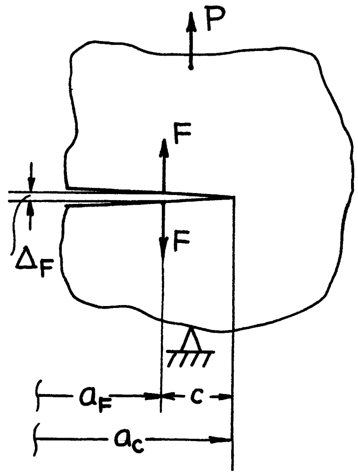


Fig. B2

Furthermore, until the crack, a, grows beyond the forces, F, it is observed that

$$\frac{\partial K_{IF}}{\partial F} = 0 \quad (a \le a_F) \tag{B13}$$

Using **Eqs. (B11), (B12), and (B13)**, in conjunction with **Eq. (B10)**, gives

$$\Delta_F = \frac{2}{E'} \int_{a_F}^{a_c} K_{IP} \frac{\partial K_{IF}}{\partial F} da \tag{B14}$$

Now both factors in the integrand are functions of crack size a and load P (but not F). By inserting formulas for K_{IP} and K_{IF} in **Eq. (B14)**, it can be directly integrated for many cases. Integration can be assisted by series or polynomial expansions in difficult cases.

Opening Displacements Near a Crack Tip

For displacements near a crack tip, that is, $c <<$ (a_F and other planar dimensions), then the form for K_{IF} is (see **page 3.6**)

$$K_{IF} = \frac{\sqrt{2}F}{\sqrt{\pi(a - a_F)}} \tag{B15}$$

The expression for K_{IP} may be replaced by a power series about a_F (or in the variable $a - a_F$)

$$K_{IP}(P, a) = K_{IP}(P, a_F) + \frac{\partial K_{IP}(P, a_F)}{\partial a} \cdot (a - a_F) + \frac{\partial^2 K_{IP}(P, a_F)}{\partial a^2} \cdot \frac{(a - a_F)^2}{2!} + \ldots \tag{B16}$$

Substituting **Eqs. (B15) and (B16)** into **Eq. (B14)** with the variable $x = a - a_F$ results in

$$\Delta_F = \frac{2\sqrt{2}}{\sqrt{\pi}E'} \int_0^c \left[K_{IP}(P, a_F) + \frac{\partial K_{IP}(P, a_F)}{\partial a} \cdot x + \frac{\partial^2 K_{IP}(P, a_F)}{\partial a^2} \cdot \frac{x^2}{2!} + \ldots \right] \frac{dx}{\sqrt{x}} \tag{B17}$$

Performing the integration

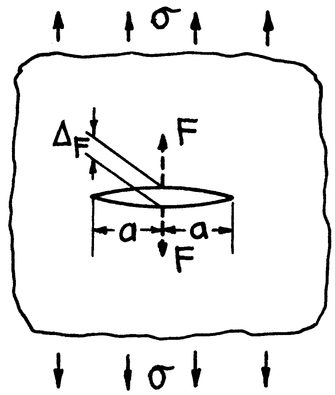
$$\Delta_F = \frac{4\sqrt{2}}{\sqrt{\pi}E'} \left[K_{IP}(P, a_F)\sqrt{c} + \frac{1}{3} \frac{\partial K_{IP}(P, a_F)}{\partial a} \cdot c^{3/2} + \ldots \right] \tag{B18}$$

If c is very small compared with a_F and other planar dimensions, then $K_{IP}(P, a_F) \cong K_{IP}(P, a_c) = K_{IP}$, and neglecting second-order terms, **Eq. (B18)** becomes

$$\Delta_F = \frac{4\sqrt{2}}{\sqrt{\pi}E'} K_{IP} \sqrt{c} \tag{B19}$$

This result, **Eq. (B19)**, is an example of the use of **Eq. (B10)** to obtain a particular result, the elastic opening of a crack near its tip. Other examples are easy to add, such as those opening displacements of cracks used for experimental purposes when measurements are made with "clip gages" (e.g., see **pages 2.4, 2.8, 2.11, 2.17, 2.20**).

Example I



For the figure shown, the K-formulas (see **pages 5.1 and 5.9**) are

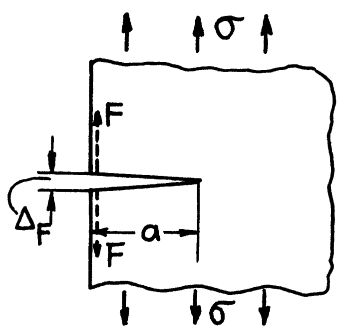
$$K_{I\sigma} = \sigma\sqrt{\pi a}$$
$$K_{IF} = \frac{F}{\sqrt{\pi a}}$$
$$K_{II} = K_{III} = 0$$

Note again that $\Delta_{no\,crack}$ is zero. Then **Eq. (B10)** reduces to

$$\Delta_F = \left\{\frac{2\sigma}{E'}\int_0^a da\right\} \times 2 = \frac{4\sigma a}{E'}$$

where the additional factor of 2 occurs because two crack tips contribute to the displacement computation.

Example II



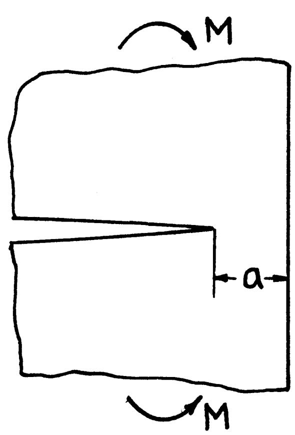
Similar to Example I (**pages 8.1 and 8.2**):

$$K_{I\sigma} = 1.1215\ \sigma\sqrt{\pi a}$$
$$K_{IF} = 1.30\frac{2F}{\sqrt{\pi a}}$$

Proceeding as before

$$\Delta_F = 5.83\frac{\sigma a}{E'}$$

Example III



For a semi-infinite plate with a semi-infinite crack leaving a neck of width a, subjected to pure moment M, applied at infinity as shown (see **page 9.1**):

$$K_{IM} = \frac{3.975\,M}{a^{3/2}}$$
$$K_{II} = K_{III} = 0$$

Applying the first form of **Eq. (B10)** to obtain the relative rotation, θ_M, of applied moments at infinity, note that $\theta_{No\,Crack}$ is zero and then

$$\theta_M = \frac{2}{E'}\int_a^\infty \frac{(3.975)^2 M}{a^3}(-da) = \frac{15.8\,M}{E'\,a^2}$$

APPENDIX C

THE WEIGHT FUNCTION METHOD FOR DETERMINING STRESS INTENSITY FACTORS[1]

Bueckner (**1970, 1971, 1972**) devised a method of determining stress intensity factor solutions,[2] which has also been discussed by **Rice (1972)**. The method depends mainly on the reciprocal theorem and other energy-method-like considerations. The method may also be extended to computations of displacements in a manner almost identical to that in **Appendix B**. For an elegant presentation of the method in all generality, the reader is refered to **Bueckner (1970, 1971, 1972) and Rice (1972)**. Here, the most important results are presented in a simplified manner.

The following analysis shows that if the complete solution (for its stress intensity factor and displacements) to a crack problem for one loading system is known, then the solution (for K) for the same cracked configuration with any other loading may be obtained directly from the known solution. To show this, consider a cracked body with loads $P_1, P_2 \ldots, P_N$ as the independently applied loads. From previous results and definitions (see **pp. 1.6 – 1.8** and **Appendixes A and B**), the Griffith energy rate is

$$\mathcal{G} = \left.\frac{\partial U}{\partial a}\right|_P = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\partial C_{ij}(a)}{\partial a}P_iP_j = \frac{1}{2}\sum_{i=1}^{N}P_i\frac{\partial u_i}{\partial a} \tag{C1}$$

where the displacements of loading points, u_i, can be written in terms of elastic compliance coefficients, $C_{ij}(a)$, as functions of crack length, a

$$u_i = \sum_{j=1}^{N} u_i^j = \sum_{j=1}^{N} C_{ij}(a)P_j \tag{C2}$$

Because of the reciprocal theorem $C_{ij} = C_{ji}$, which was used in writing the above equations.

On the other hand, the Griffith energy rate may be written in terms of stress-intensity factors (see **p. 1.8**) as

$$\mathcal{G} = \frac{K^2}{E'} = \frac{1}{E'}\sum_{i=1}^{N}\sum_{j=1}^{N}k_i(a)k_j(a)P_iP_j \tag{C3}$$

---

[1] **Paris 1976.**

[2] **For earlier "Green's Function" from concentrated force solutions with results similar to weight functions, see Paris (1957, 1960), Barenblatt (1962), and Sih (1962b).**

since the stress intensity factor, K, is linearly dependent on the loads, P_i, or

$$K = \sum_{i=1}^{N} K_i = \sum_{i=1}^{N} k_i(a)P_i \tag{C4}$$

Equating the double sums in both results for $\mathcal{G}$ above,[3] that is, (**C1**) **and** (**C4**), and noting that since this must be true for any values of the loads, $P_1, P_2, \ldots, P_N$, the coefficients must be identical, term by term, then

$$\frac{k_i(a)k_j(a)}{E'} = \frac{1}{2}\frac{\partial C_{ij}(a)}{\partial a} \tag{C5}$$

Let a full solution be known for just one of the loads, say P_m. Then, rearranging the latest result

$$k_i(a) = \frac{E'}{2}\frac{\partial C_{im}(a)}{\partial a}\frac{1}{k_m(a)}$$

or from **Eq. (C4)**

$$k_i(a) = \frac{E'}{2}\frac{\partial C_{im}(a)}{\partial a}\frac{P_m}{K_m} \tag{C6}$$

By saying the solution is known for a load, P_m, it means that K_m is known and C_{im} is known, since the displacements, u_i^m, for the load at P_m are presumed to be known and from **Eq. (C2)**

$$C_{im} = \frac{u_i^m}{P_m}$$

Combining (**C4**) **and** (**C6**)

$$K = \sum_{i=1}^{N} k_i(a)P_i = \frac{E'}{2}\frac{P_m}{K_m}\sum_{i=1}^{N}\frac{\partial C_{im}(a)}{\partial a}P_i \tag{C7}$$

Thus K can be found for loads, P_i, from results obtained from just one load, P_m. This is the desired result.

For arbitrary distributed tractions, $T(s)$, over a surface, s, instead of discrete forces, P_i, the form of the result (**C7**) becomes

$$K = \int f_m(s,a)T(s)ds \tag{C8}$$

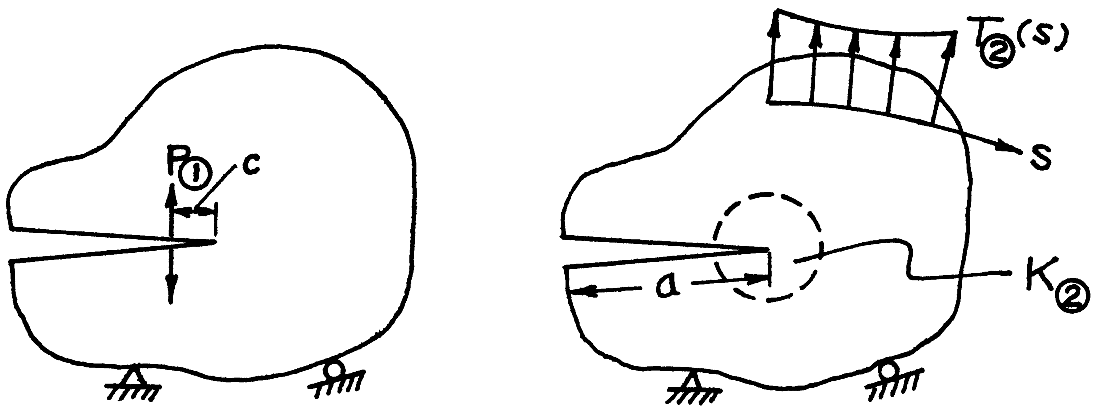
where $f_m(s,a)$ is the "weight function" as determined entirely from the solution for a load (or loading system) characterized by "m." For this result, note that

$$f_m(s,a) = \frac{E'}{2K_m^{(a)}} \cdot \frac{\partial u^m(s,a)}{\partial a} \tag{C9}$$

where $u^m(s,a)$ are displacements at s in the direction of the tractions $T(s)$ but caused only by the loading system characterized by "m."

[3] **As suggested by J. R. Rice, private communication, 1974.**

A Special Method of Determining the Weight Function — Mode I



State ①
(Mode I)

State ②

Fig. C1

In a two-dimensional problem, let loading state ① be the known loading state (corresponding to "m" above) where concentrated forces, $P_{①}$, are applied on the crack surface at a distance, c, from the crack tip. Let loading state ②(for the same configuration) be one of arbitrary tractions for which it is desired to determine $K_{②}$. By reciprocal theorem:

$$P_{①}\, u_{②} = \int T_{②} u_{①}\, ds \tag{C10}$$

where the displacements, u, form reciprocal work products (e.g., $u_{②}$ is the displacement at the location of $P_{①}$ in the direction of $P_{①}$ but due to loading state ②).

Now, presume that the distance, c, from the crack tip to the loading forces in state ① is very small (i.e., approaching zero compared with other dimensions). Then the displacement $u_{②}$ will be within the crack-tip stress field for state ② or (see **p. 1.2**, **Eq. (1)**, etc.)

$$u_{②} = 2v\Big|_{\substack{r=c\\ \theta=\pi}} = \frac{4\sqrt{2}}{\sqrt{\pi}E'} K_{I②}\sqrt{c} \tag{C11}$$

Note that, due to the symmetrical (with respect to the crack) force system selected for state ①, only Mode I fields contribute work-producing displacements, $u_{②}$, thus the K being computed here is only the Mode I component. Substituting (**C11**) and rearranging (**C10**) gives

$$K_{I②} = \frac{\sqrt{\pi}E'}{4\sqrt{2}} \frac{1}{P_{①}\sqrt{c}} \int T_{②} u_{①}\, ds \tag{C12}$$

As c diminishes to zero, let $P_{①}\sqrt{c}$ remain a finite constant, choosing for later convenience

$$\frac{P_{①}\sqrt{c}}{\pi} = B_I \tag{C13}$$

Considering the results on page 3.6, this leads to a local situation in state ① of

$$Z_{I①}(z) = B_I \Big/ z^{3/2} \tag{C14}$$

As c approaches zero, the boundaries by comparison become infinitely far away. Again, substituting (**C13**) into (**C12**) gives

$$K_{I②} = \frac{E'}{4\sqrt{2\pi}B_I}\int T_{②}u^I_{①}ds \tag{15}$$

where $T_{②}$ are the applied tractions and $u^I_{①}$ are now the corresponding displacements without applied loads but with insertion of a local Mode I singularity of $z^{-3/2}$ type [4] of strength "B_I" as in (**C14**), at the crack tip where $K_{②}$ is desired. The weight function is then

$$f_{\mathrm{Im}}(a,s) = \frac{E'u^I_{①}(a,s)}{4\sqrt{2\pi}B_I} \tag{C16}$$

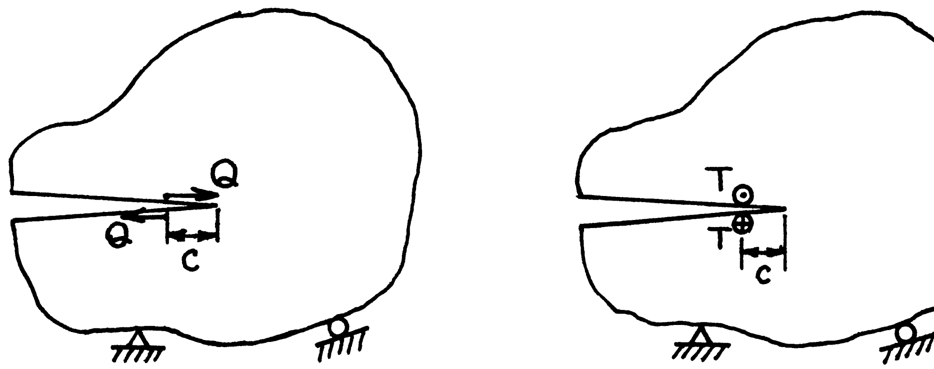
Bueckner (**1970, 1971, 1972**) obtained a similar result by a less direct approach.

It is easy to visualize inserting the local singularity described by (**C14**) into a finite element scheme to determine resulting displacements, $u^I_{①}(a,s)$, for all mesh points with no other loads present. This has distinct advantages over other methods, as the solution generated applies to <u>all</u> possible loadings. Other results are possible using collocation or direct solutions using this method.

This derivation considers Mode I only, and the resulting stress intensity factor in (**C15**) is of a Mode I type. However, it is possible to replace state ① with its Mode II or Mode III counterparts (see **p. 3.6**) and rederive results for K_{II} and K_{III}, as well as K_I, as follows.

[4] **Let the $z^{-3/2}$ singularity be known as the "Bueckner Type," with its strength appropriately denoted as "B."**

Mode II and Mode III Weight Functions



State ①
(Mode II)
(a)

State ①
(Mode III)
(b)

Fig. C2

By repeating the derivation of the preceding section, that is, (**C10**) through (**C16**), but replacing state ①in **Fig. C1** with its Mode II or Mode III counterpart, as shown in **Fig. C2 (a) and (b)**, then weight functions may be developed directly for Mode II and Mode III stress intensity factors. The results are as follows.

Mode II

$$K_{II②} = \frac{E'}{4\sqrt{2\pi}B_{II}} \int T_{②} u_{①}^{II} ds \qquad (\mathbf{C17})$$

where $u_{①}^{II}$ is now caused by inserting a local field of the Bueckner type at the crack tip of interest, that is,

$$Z_{II①}(z) = B_{II} \Big/ z^{3/2} \qquad (\mathbf{C18})$$

Mode III

$$K_{III②} = \frac{G}{2\sqrt{2\pi}B_{III}} \int T_{②} u_{①}^{III} ds \qquad (\mathbf{C19})$$

where $u_{①}^{III}$ is caused by inserting the Bueckner singularity

$$Z_{III①}(z) = B_{III} \Big/ z^{3/2} \qquad (\mathbf{C20})$$

Therefore it is equally easy to insert Mode II and Mode III singularities in problems to obtain the Mode II and Mode III weight functions

$$f_{IIm}(a,s) = \frac{E' u_{①}^{II}(a,s)}{4\sqrt{2\pi}B_{II}} \qquad (\mathbf{C21})$$

and

$$f_{IIIm}(a,s) = \frac{G\,u_{①}^{III}(a,s)}{2\sqrt{2\pi}B_{III}} \tag{C22}$$

where now **Eq. (C8)** may be applied individually for each of the three modes of stress intensity factors.

Near Tip Bueckner Displacement Fields

The displacement fields may be computed for Bueckner-type singularities, that is [see **Eqs. (C14), (C18) and (C20)**],

$$Z(z) = B \Big/ z^{3/2}$$

for each mode, making use of the usual r, θ coordinates (see **Fig. 2, p. 1.2**) and **Eqs. (39), (56), or (59)** as is appropriate for each mode. The results (for plane strain for Modes I and II) are as follows.

Mode I

$$\begin{aligned} u &= \frac{B_I}{G\sqrt{r}}\cos\frac{\theta}{2}\left[2\nu - 1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] \\ \nu &= \frac{B_I}{G\sqrt{r}}\sin\frac{\theta}{2}\left[2 - 2\nu - \cos\frac{\theta}{2}\cos\frac{3\theta}{2}\right] \\ w &= 0 \end{aligned} \tag{C23}$$

Mode II

$$\begin{aligned} u &= \frac{B_{II}}{G\sqrt{r}}\sin\frac{\theta}{2}\left[2 - 2\nu + \cos\frac{\theta}{2}\cos\frac{3\theta}{2}\right] \\ \nu &= \frac{B_{II}}{G\sqrt{r}}\cos\frac{\theta}{2}\left[1 - 2\nu + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] \\ w &= 0 \end{aligned} \tag{C24}$$

Mode III

$$\begin{aligned} u &= 0 \\ \nu &= 0 \\ w &= \frac{B_{III}}{G\sqrt{r}}2\sin\frac{\theta}{2} \end{aligned} \tag{C25}$$

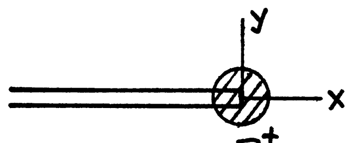
For plane stress for Modes I and II, replace ν by $\nu/(1+\nu)$ and also note that $w \neq 0$.

For generating weight functions, these fields should be inserted locally at the crack tip where K is desired. They are then the actual weight function displacements that should be used for loads near the crack tip.

For a semi-infinite crack in an infinite body, these fields are the weight function displacements for the whole problem (two-dimensional problems). Some examples follow, in which these results are used for very simple problems so as to illustrate the method. The power of the method, however, is only fully appreciated with more complicated problems, when combined with numerical, finite element, and other procedures.

Closed-Form Weight Functions

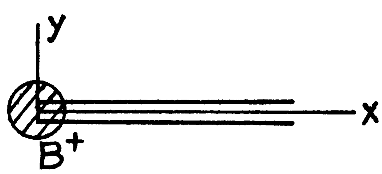
Closed forms for Westergaard stress functions, Z and $\overline{Z}$, can be written to form weight function displacement fields throughout a body. The technique of finding such stress functions is much the same as for normal crack stress analysis problems, except that the crack tip for which the weight function is desired will have a $z^{-3/2}$ (in Z) of strength, B_I. Some examples are as follows (each applies to all three modes, I, II, III):



$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \begin{Bmatrix} B_I \\ B_{II} \\ B_{III} \end{Bmatrix} \frac{1}{z^{3/2}}$$

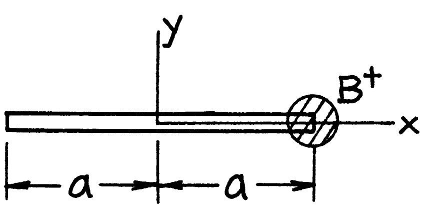
$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = -\begin{Bmatrix} B_I \\ B_{II} \\ B_{III} \end{Bmatrix} \frac{2}{z^{1/2}}$$

NOTE: All Z and $\overline{Z}$ expressions below apply to all three modes as indicated here.



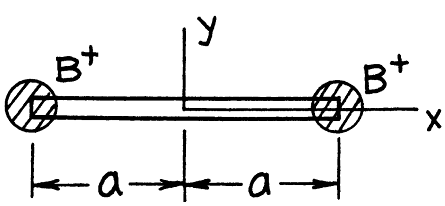
$$Z(z) = \frac{iB}{z^{3/2}}$$

$$\overline{Z}(z) = -\frac{2iB}{z^{1/2}}$$



$$Z(z) = \frac{B\sqrt{2a}}{(z-a)^{3/2}(z+a)^{1/2}}$$

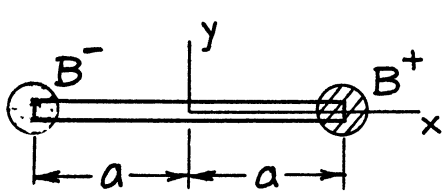
$$\overline{Z}(z) = -B\sqrt{\frac{2}{a}}\left(\frac{z+a}{z-a}\right)^{1/2}$$



symmetric about y-axis

$$Z(z) = \frac{2B\sqrt{2a}\,z}{\left(z^2-a^2\right)^{3/2}}$$

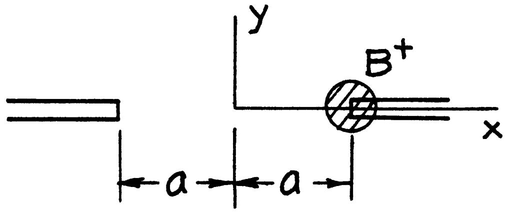
$$\overline{Z}(z) = \frac{-2B\sqrt{2a}}{\left(z^2-a^2\right)^{1/2}}$$



skew-symmetric about y-axis

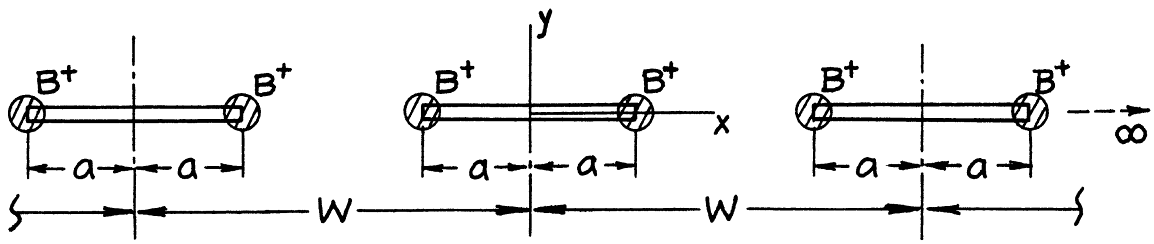
$$Z(z) = \frac{B(2a)^{3/2}}{\left(z^2-a^2\right)^{3/2}}$$

$$\overline{Z}(z) = \frac{-2B\sqrt{2a}}{\left(z^2-a^2\right)^{1/2}} \cdot \frac{z}{a}$$



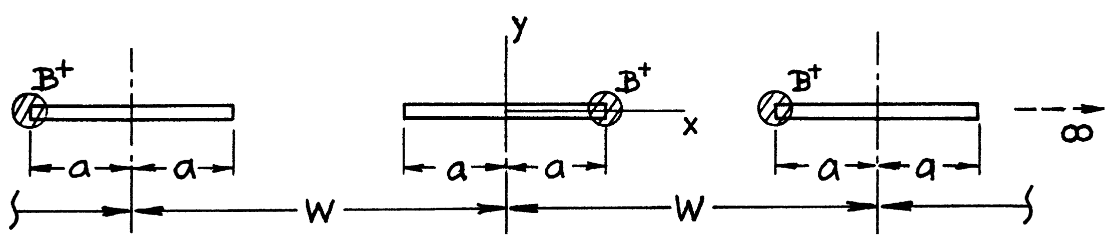
$$Z(z) = \frac{B\sqrt{2a}}{(a-z)^{3/2}(a+z)^{1/2}}$$

$$\overline{Z}(z) = B\sqrt{\frac{2}{a}}\left(\frac{a+z}{a-z}\right)^{1/2}$$



$$Z(z) = \frac{2B\left(\frac{\pi}{W}\cos\frac{\pi a}{W}\right)^{3/2}\left(2\sin\frac{\pi a}{W}\right)^{1/2}\sin\frac{\pi z}{W}}{\left(\sin\frac{\pi z}{W}-\sin\frac{\pi a}{W}\right)^{3/2}\left(\sin\frac{\pi z}{W}+\sin\frac{\pi a}{W}\right)^{3/2}}$$

$$\overline{Z}(z) = \frac{-2B\left(\frac{2\pi}{W}\tan\frac{\pi a}{W}\right)^{1/2}\cos\frac{\pi z}{W}}{\left[\left(\sin\frac{\pi z}{W}\right)^2-\left(\sin\frac{\pi a}{W}\right)^2\right]^{1/2}}$$



$$Z(z) = \frac{B\left(\frac{\pi}{W}\cos\frac{\pi a}{W}\right)^{3/2}\left(2\sin\frac{\pi a}{W}\right)^{1/2}}{\left(\sin\frac{\pi z}{W}-\sin\frac{\pi a}{W}\right)^{3/2}\left(\sin\frac{\pi z}{W}+\sin\frac{\pi a}{W}\right)^{1/2}}$$

$$\overline{Z}(z) = B\left\{\frac{-\left(\frac{2\pi}{W}\tan\frac{\pi a}{W}\right)^{1/2}\cos\frac{\pi z}{W}}{\left[\left(\sin\frac{\pi z}{W}\right)^2-\left(\sin\frac{\pi a}{W}\right)^2\right]^{1/2}} + i\ \cot\frac{\pi a}{W}\left(\frac{\pi}{W}\sin\frac{2\pi a}{W}\right)^{1/2}\Pi_c\left[\sin^{-1}\left(\frac{\sin\frac{\pi z}{W}}{\sin\frac{\pi a}{W}}\right), 1, \sin\frac{\pi a}{W}\right]\right\}$$

$$\Pi_c[\varphi, n, k] = \int\frac{d\varphi}{\left(1-n\ \sin^2\varphi\right)\left(1-k^2\ \sin^2\varphi\right)^{1/2}}$$

An Example of Finite Element Results

As mentioned earlier, weight function displacements can be obtained using finite element analysis by putting a small hole at the crack tip and inserting the Bueckner-type field as boundary conditions on the hole, that is, by using **Eqs. (C23) [or (C24) or (C25)]** as boundary conditions on the hole, with all other surfaces stress free, in order to determine the state ① displacements, $u_{①}$.

As an example, consider the single edge cracked strip for which a variety of loadings are avaliable with previously tabulated results (see **Paris 1975a** and **pages 2.10, 2.13, 2.16, 2.27, 2.29**, etc.). The particular strip selected here (see the figure on **page 2.10**) is $a = b/2$, $2h = 6b$, with a circular hole at the crack of radius $r = .006944b$ and $\nu = 0.3$. A mesh of 352 elements and 398 nodes was used, as shown in Fig. C3 (with the inner circles of elements removed and enlarged for clarity).

Table C1 gives results obtained by simply inserting the displacement field, **Eqs. (C23)**, at 25 points on the upper half of the hole, and constraining points on the crack plane ahead of the crack to remain on that plane. As implied, only one-half of the problem requires treatment. The points on the table are numbered as indicated on the mesh in Fig. C3 and located by coordinates x/b and y/b. The corresponding components of the displacements, $u_{①}$, for use in the integral to obtain $K_{I②}$, are

$$\bar{u}/b = \frac{E'\left(u_{①}\right)_x}{2\sqrt{2\pi}B_I b}$$

$$\bar{v}/b = \frac{E'\left(u_{①}\right)_y}{2\sqrt{2\pi}B_I b}$$

Thus $K_{I②}$ may be obtained from components of tractions for one-half of a problem (with respect to crack-plane symmetry) from

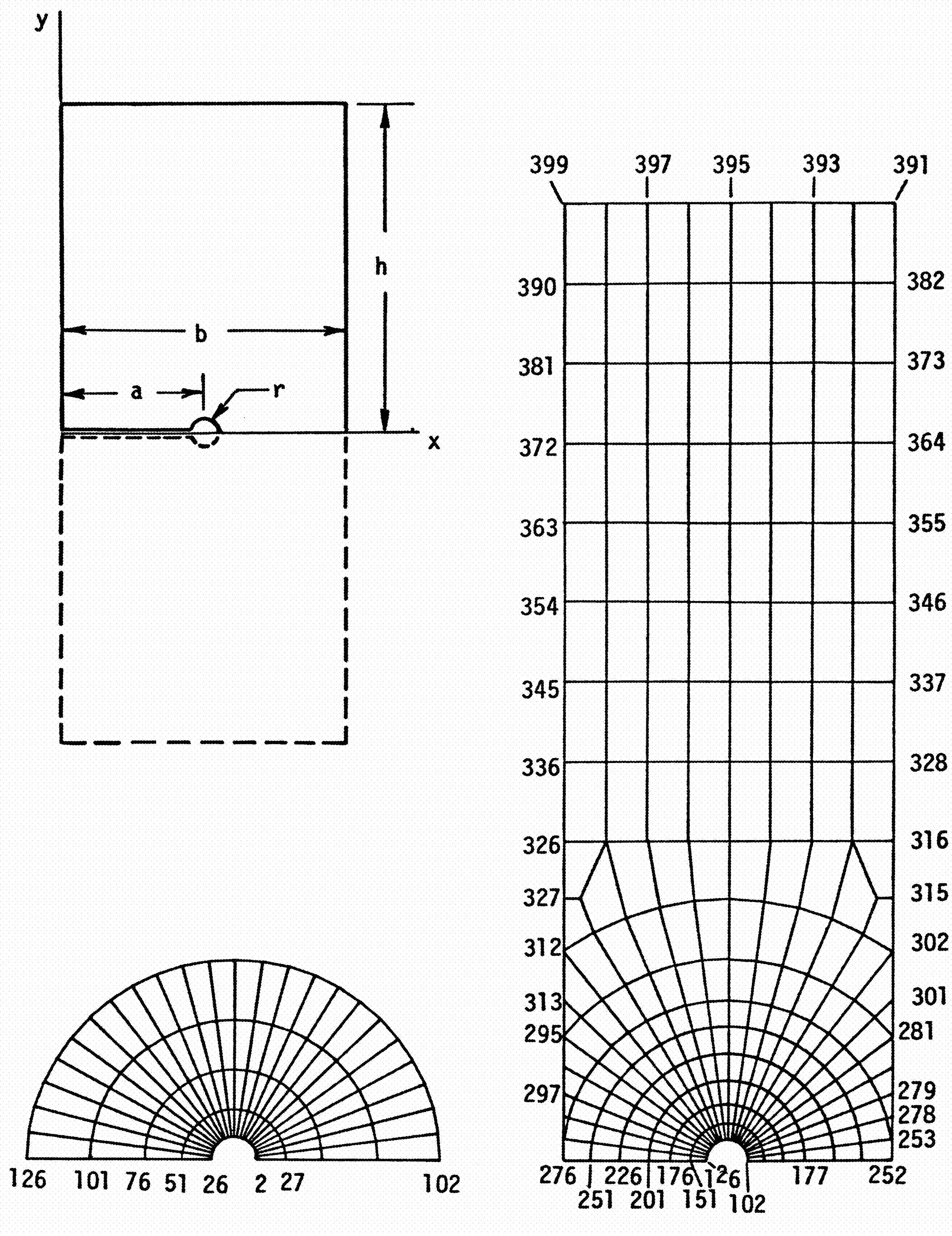
$$K_{I②} = \int (T_x\bar{u} + T_y\bar{v})ds$$

Using this formula and the table, the following accuracies are observed in comparative results for various loadings:

a. Uniform tension applied at the ends — 2.9% (see **p. 2.10**)
b. Uniform pressure on the crack surface — <4.9%
c. Pure bending (moment applied anywhere beyond $y/b = 1.5$) — 3.2% (see **p. 2.13**)
d. Four-point bending (with loads at $y/b = 1$ and 3) — 3.2%
e. Three-point bending (with a span of $4b$) — 3.6% (see **p. 2.16**)

This is only an example and better accuracies may be produced by finer finite element meshes. However, the table allows the reader to calculate K_I results of comparable accuracies for any loading for this configuration. Its advantages and generality are apparent.

The same method applies to Modes II and III simply by using the proper fields, **Eqs. (C24) and (C25)**. Moreover, it is also applicable to residual stress, thermal stress, and body force problems via simple superposition or direct application. The method is also applicable to three-dimensional finite element analysis.


y
h
b
a
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327
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312
302
313
301
295
281
297
279
278
253
276
226
176
126
177
252
251
201
151
102
126
101
76
51
26
2
27
102

Fig. C3

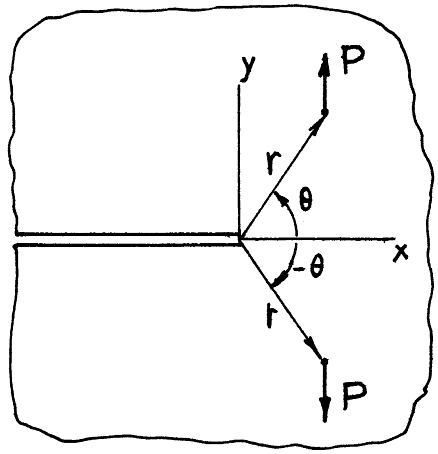
Weight Function Displacements for an Edge Cracked Strip

$$\left(\frac{a}{b} = 0.5,\ \frac{h}{b} = 3,\ \nu = 0.3,\ \frac{r}{b} = .006944\right)$$

Point	x/b	y/b	$\bar{u}/b$	$\bar{v}/b$
2	0.5069	0	- 2.736	0
27	0.5156	0	- 1.767	0
102	0.5625	0	- 0.595	0
177	0.75	0	0.285	0
26	0.4931	0	- 0.000003	9.575
51	0.4844	0	- 0.232	7.066
76	0.4722	0	- 0.269	5.930
101	0.4566	0	- 0.297	5.395
126	0.4375	0	- 0.319	5.187
151	0.3888	0	- 0.319	5.229
176	0.3264	0	- 0.346	5.670
201	0.25	0	- 0.368	6.374
226	0.1666	0	- 0.385	7.214
251	0.0833	0	- 0.393	8.084
276	0	0	- 0.396	8.960
297	0	0.207	1.795	8.971
295	0	0.3837	3.719	8.956
313	0	0.5	4.988	8.933
312	0	0.6516	6.634	8.908
327	0	0.820	8.453	8.890
326	0	1.00	10.401	8.887
336	0	1.25	13.116	8.887
345	0	1.50	15.838	8.889
354	0	1.75	18.563	8.891
363	0	2.00	21.289	8.892
372	0	2.25	24.015	8.892
381	0	2.50	26.740	8.892
390	0	2.75	29.466	8.892
399	0	3.00	32.191	8.892
397	0.25	3.00	32.191	6.166
395	0.50	3.00	32.191	3.441
393	0.75	3.00	32.191	0.715
391	1.0	3.00	32.191	-2.010
382	1.0	2.75	29.466	-2.010
373	1.0	2.50	26.740	-2.010
364	1.0	2.25	24.015	-2.010
355	1.0	2.00	21.289	-2.010
346	1.0	1.75	18.564	-2.011
337	1.0	1.50	15.838	-2.012
328	1.0	1.25	13.109	-2.015
316	1.0	1.0	10.371	-2.017
315	1.0	0.820	8.386	-2.015
302	1.0	0.6516	6.515	-1.980
301	1.0	0.50	4.819	-1.912
281	1.0	0.3837	3.525	-1.744
279	1.0	0.207	1.774	-1.249
278	1.0	0.134	1.228	-0.865
253	1.0	0.0653	0.904	-0.437
252	1.0	0	0.805	0

Closed-Form Examples

Example (1)



Consider an infinite sheet with a semi-infinite crack subjected to a pair of equal and opposite forces (per unit thickness), P, as shown. The stress solution with a Bueckner-type Mode I singularity (and no loading forces) is

$$Z_{I①}^{(z)} = \frac{B_I}{z^{3/2}} \text{(Any } z)$$

The stress intensity factor for the loads shown may be computed from

$$K_I = \frac{E'}{4\sqrt{2\pi}B_I} \int T_② u^I_① ds$$

Now because of x-axis symmetry

$$\int T_② u^I_① ds = 2P \cdot v_①(r, \theta)$$

From **Eqs. (39)**, for plane strain conditions,

$$2Gv_① = 2(1-\nu)\ \mathrm{Im}\overline{Z}_{I①} - y\ \mathrm{Re}Z_{I①}$$

where for plane stress, ν can be replaced by $\nu/(1+\nu)$. Substituting $Z_{I①}$ and taking $z = re^{i\theta}$, etc. [see also **Eqs. (C23)**]

$$v_① = \frac{B_I}{G\sqrt{r}} \sin\frac{\theta}{2}\left[2(1-\nu) - \cos\frac{\theta}{2}\cos\frac{3\theta}{2}\right]$$

Continuing the substitutions

$$K_I = \frac{P}{(1-\nu)\sqrt{2\pi r}} \sin\frac{\theta}{2}\left[2(1-\nu) - \cos\frac{\theta}{2}\cos\frac{3\theta}{2}\right]$$

This result is a somewhat simpler form of the solution of **page 3.4**.

For the special case $r = b$ and $\theta = \pi$, the result is the same as **page 3.6**, or

$$K_I = \frac{2P}{\sqrt{2\pi b}}$$

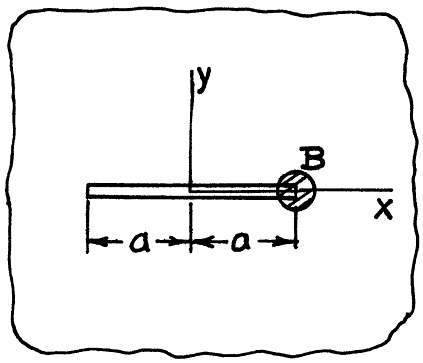
and for the special case $r = y_0$ and $\theta = \frac{\pi}{2}$, the result is the same as **page 3.5**, or

$$K_I = \frac{P}{\sqrt{\pi y_0}}\left[\frac{5-4\nu}{4-4\nu}\right]$$

Therefore in this case getting the K_I through the weight function method gives direct algebraic access to a simpler form for some uses.

Note also from (C24) and (C25) that no Mode II or Mode III work is done by the forces, P, with these displacements. Thus $K_{II} = K_{III} = 0$.

Example (2)



Consider a finite crack of length 2a, with a Bueckner-type singularity at one end. The stress function is [as verified from **page 5.9** and **Eq. (C13)**]

$$Z_{I①}^{(z)} = \frac{B_I\sqrt{2a}}{(z-a)^{3/2}(z+a)^{1/2}}$$

Consider now the problem of uniform stress, σ, parallel to the y-axis at infinity. Then

$$\int T_{②} u_{①}^{I}\, ds = 2\int_{-\infty}^{\infty} \sigma v \Big|_{y\gg a} dx$$

At great distances from the crack $(y>>a)$

$$Z_{I①\infty}^{(z)} \to \frac{B_I\sqrt{2a}}{z^2}$$

and integrating

$$\overline{Z}_{I①\infty}^{(z)} \to -\frac{B_I\sqrt{2a}}{z}$$

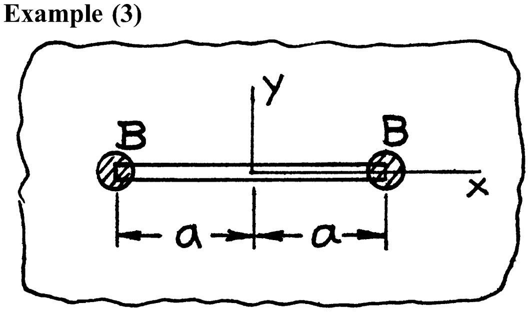
Substituting these expressions for Z and $\overline{Z}$ into the expression for ν [**Eqs. (39)**, **page 1.13**] gives

$$\int T_{②} u_{①}^{I}\, ds = \frac{4\pi\sigma B_I\sqrt{2a}}{E'}$$

Thus from **Eq. (C15)**

$$K_I = \sigma\sqrt{\pi a}$$

Example (3)



For problems like Example (2), with y-axis symmetry, it is permitted to insert Bueckner singularities at both crack tips or

$$Z_{I①}^{(z)} = \frac{B_I\sqrt{2a}\,2z}{\left(z^2-a^2\right)^{3/2}}$$

Note that for $z>>a$, the stress function becomes twice the result in the previous example, that is,

$$Z_{I①\infty}^{(z)} \to \frac{2B_I\sqrt{2a}}{z^2}(z\gg a)$$

where the analysis would follow in the same manner for $2K_I$ instead of K_I, as two crack tips are simultaneously considered.

However, here it is very easy to integrate $Z_{I①}$ (for any z) to obtain

$$\overline{Z}^{(z)}_{I①} = -\frac{2B_I\sqrt{2a}}{\left(z^2 - a^2\right)^{1/2}} \qquad (\text{Any } z)$$

Hence it is possible to compute the weight function displacements easily for any point.

As a special case consider the displacement only on the crack surface, that is,

$$2Gv = 2(1-\nu)\ \mathrm{Im}\overline{Z}_I + 0 \quad (y = 0, |x| < a)$$

$$2Gu = 0$$

or, substituting for $\overline{Z}$

$$v = \frac{2(1-\nu)B_I\sqrt{2a}}{G}\left(a^2 - x^2\right)^{-1/2}$$

For a crack surface loaded by compressive stresses symmetric with respect to both axes, $\sigma(x) = \sigma(-x)$, then

$$2K_I = \frac{E'}{4\sqrt{2\pi}B_I}\int_0^a 2\sigma(x)\cdot 2v(x)dx$$

which upon substitution gives

$$K_I = \frac{2\sqrt{a}}{\sqrt{\pi}}\int_0^a \frac{\sigma(x)}{\left(a^2 - x^2\right)^{1/2}}dx$$

and

$$K_{II} = 0$$

which are well known results.

As another application consider equal and opposite forces P in the y direction at the origin, or

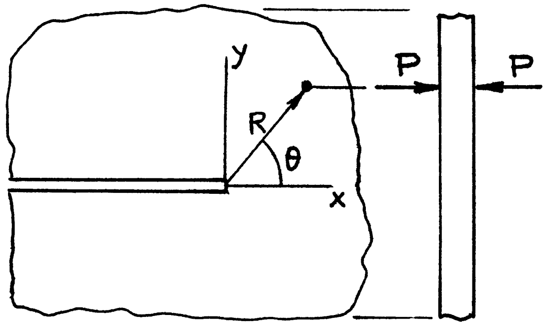
$$2K_I = \frac{E'}{4\sqrt{2\pi}B_I}P\cdot 2v\Big|_{\substack{x=0\\y=0}}$$

which gives

$$K_I = \frac{P}{\sqrt{\pi a}}$$

A result that is easily checked by **page 5.9**

Example (4)



Consider a crack tip in a sheet that is subject to pinching forces P as shown. The displacement per unit thickness is

$$u_{①} = \varepsilon_z = \frac{-\nu}{E}(\sigma_x + \sigma_y)$$

which from **Eqs. (38) and (55)** becomes

Mode I

$$u_{①} = -\frac{2\nu}{E}\ \mathrm{Re}Z_I$$

Mode II

$$u_{①} = -\frac{2\nu}{E}\ \mathrm{Im}Z_{II}$$

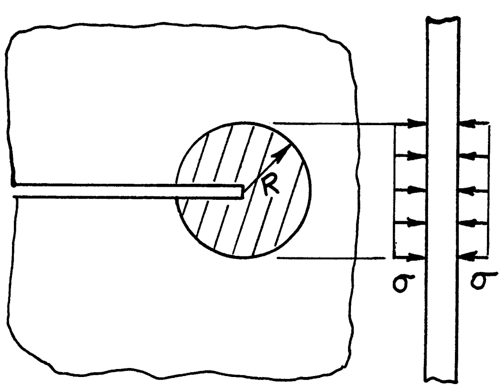
and for both modes

$$Z(z) = \frac{B}{z^{3/2}} = \frac{B}{r^{3/2}}\left(\cos\frac{3\theta}{2} - i\ \sin\frac{3\theta}{2}\right)$$

where substituting these results into (C15) and (C17) gives

$$K_I = \frac{\nu P}{2\sqrt{2\pi}R^{3/2}}\cos\frac{3\theta}{2}$$

$$K_{II} = \frac{-\nu P}{2\sqrt{2\pi}R^{3/2}}\sin\frac{3\theta}{2}$$



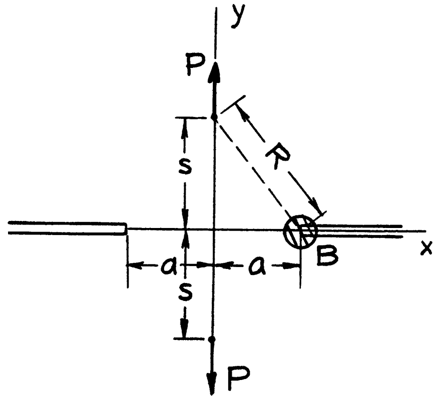
These expressions can be used to obtain results for a crack tip centered in a zone of radius R and pinching stress σ to give

$$K_I = \frac{-4\nu}{3\sqrt{2\pi}}\sigma\sqrt{R}$$

$$K_{II} = 0$$

Many other solutions for pinching load problems are compiled in **Appendix E**.

Example (5)



Consider two opposing semi-infinite cracks in a plate leaving a neck of width $2a$, loaded by opposing concentrated forces P, each centered on the neck at a distance s from the plane of the neck. The stress function solution with a Bueckner-type-singularity at the right-hand crack tip is

$$Z(z) = \frac{B\sqrt{2a}}{(a-z)^{3/2}(a+z)^{1/2}}$$

$$\overline{Z}(z) = \int Z(z)dz = B\sqrt{\frac{2}{a}}\left(\frac{a+z}{a-z}\right)^{1/2}$$

Now from **Eq. (C15)**

$$K = \frac{E'}{4\sqrt{2\pi}B}(2Pv)$$

where v is the displacement at the load point to be computed using the above Z and $\overline{Z}$ in **Eq. (39)** or

$$2\,Gv = 2(1-\nu)\ \mathrm{Im}\overline{Z} - y\ \mathrm{Re}Z$$

The calculation could be carried out for any position of the loading forces (not necessarily on the y-axis), but the algebra becomes messy. Thus for the location of loads as shown in the figure

$$\mathrm{Im}\overline{Z} = B\sqrt{\frac{2}{a}}\frac{s}{\left(a^2+s^2\right)^{1/2}}$$

and

$$\mathrm{Re}Z = \frac{B\sqrt{2a}\,a}{\left(a^2+s^2\right)^{3/2}}$$

Defining the distance between a load point and a crack tip as R or $R^2 = a^2 + s^2$, and combining expressions

$$K = \frac{P}{\sqrt{\pi a}}\left[\frac{s}{R} + \frac{(1+\nu)}{2}\frac{a^2 s}{R^3}\right]$$

This result is noted as a simpler way to obtain K compared with the solution on **page 4.3** (and **pages 4.9, 4.11**, etc.).

Appendix D

Anisotropic Linear-elastic Crack-tip Stress Fields

Hooke's law for a homogeneous (rectilinear) anisotropic material is

$$\left.\begin{aligned}
\varepsilon_x &= a_{11}\sigma_x + a_{12}\sigma_y + a_{13}\sigma_z + a_{14}\tau_{yz} + a_{15}\tau_{zx} + a_{16}\tau_{xy}\\
\varepsilon_y &= a_{21}\sigma_x + \cdots\\
\varepsilon_z &= a_{31}\sigma_x + \cdots\\
\gamma_{yz} &= a_{41}\sigma_x + \cdots\\
\gamma_{zx} &= a_{51}\sigma_x + \cdots\\
\gamma_{xy} &= a_{61}\sigma_x + a_{62}\sigma_y + a_{63}\sigma_z + a_{64}\tau_{yz} + a_{65}\tau_{zx} + a_{66}\tau_{xy}
\end{aligned}\right\} \tag{D1}$$

where, from reciprocity, $a_{ij} = a_{ji}$.

The generalized Hooke's law may be reduced by imposing conditions of plane strain (or stress) and pure shear to lead to tractable problems of determining crack-tip stress fields. Proceeding as in **Sih (1965b) or Paris (1965)**, the crack-tip stress field equations can be written as follows.

<u>For plane strain</u>

$$\left.\begin{aligned}
\sigma_x &= \frac{K_{Ia}}{\sqrt{2\pi r}}\,\mathrm{Re}\left\{\frac{\mu_1\mu_2}{\mu_1-\mu_2}\left(\frac{\mu_2}{\sqrt{\cos\theta+\mu_2\sin\theta}}-\frac{\mu_1}{\sqrt{\cos\theta+\mu_1\sin\theta}}\right)\right\}\\
&\quad+\frac{K_{IIa}}{\sqrt{2\pi r}}\,\mathrm{Re}\left\{\frac{1}{\mu_1-\mu_2}\left(\frac{\mu_2^2}{\sqrt{\cos\theta+\mu_2\sin\theta}}-\frac{\mu_1^2}{\sqrt{\cos\theta+\mu_1\sin\theta}}\right)\right\}\\
\sigma_y &= \frac{K_{Ia}}{\sqrt{2\pi r}}\,\mathrm{Re}\left\{\frac{1}{\mu_1-\mu_2}\left(\frac{\mu_1}{\sqrt{\cos\theta+\mu_2\sin\theta}}-\frac{\mu_2}{\sqrt{\cos\theta+\mu_1\sin\theta}}\right)\right\}\\
&\quad+\frac{K_{IIa}}{\sqrt{2\pi r}}\,\mathrm{Re}\left\{\frac{1}{\mu_1-\mu_2}\left(\frac{1}{\sqrt{\cos\theta+\mu_2\sin\theta}}-\frac{1}{\sqrt{\cos\theta+\mu_1\sin\theta}}\right)\right\}\\
\tau_{xy} &= \frac{K_{Ia}}{\sqrt{2\pi r}}\,\mathrm{Re}\left\{\frac{\mu_1\mu_2}{\mu_1-\mu_2}\left(\frac{1}{\sqrt{\cos\theta+\mu_1\sin\theta}}-\frac{1}{\sqrt{\cos\theta+\mu_2\sin\theta}}\right)\right\}\\
&\quad+\frac{K_{IIa}}{\sqrt{2\pi r}}\,\mathrm{Re}\left\{\frac{1}{\mu_1-\mu_2}\left(\frac{\mu_1}{\sqrt{\cos\theta+\mu_1\sin\theta}}-\frac{\mu_2}{\sqrt{\cos\theta+\mu_2\sin\theta}}\right)\right\}
\end{aligned}\right\} \tag{D2}$$

where (other corresponding components of stress and displacement omitted) μ_1 and μ_2 are from each of the conjugate pairs of roots of

$$A_{11}\mu^4 - 2A_{16}\mu^3 + (2A_{12} + A_{66})\mu^2 - 2A_{26}\mu + A_{22} = 0 \tag{D3}$$

and where A_{ij} are obtained by using the restrictions on strain to eliminate the appearance of z-components of stress to give

$$\left.\begin{aligned} \varepsilon_x &= A_{11}\sigma_x + A_{12}\sigma_y + A_{16}\tau_{xy} \\ \varepsilon_y &= A_{21}\sigma_x + A_{22}\sigma_y + A_{26}\tau_{xy} \\ \gamma_{xy} &= A_{61}\sigma_x + A_{62}\sigma_y + A_{66}\tau_{xy} \end{aligned}\right\} \tag{D4}$$

where again, $A_{ij} = A_{ji}$.

<u>For pure shear</u>

$$\left.\begin{aligned} \tau_{yz} &= \frac{K_{IIIa}}{\sqrt{2\pi r}}\ \mathrm{Re}\left\{\frac{1}{\sqrt{\cos\theta+\mu_5\ \sin\theta}}\right\} \\ \tau_{xz} &= \frac{-K_{IIIa}}{\sqrt{2\pi r}}\ \mathrm{Re}\left\{\frac{\mu_5}{\sqrt{\cos\theta+\mu_5\ \sin\theta}}\right\} \end{aligned}\right\} \tag{D5}$$

where μ_5 is one of the conjugate roots of

$$A_{55}\mu^2 - 2A_{45}\mu + A_{44} = 0 \tag{D6}$$

and where A_{ij} are obtained by using the restrictions for pure shear to give

$$\left.\begin{aligned} \gamma_{yz} &= A_{44}\tau_{yz} + A_{45}\tau_{xz} \\ \gamma_{xz} &= A_{45}\tau_{yz} + A_{55}\tau_{xz} \end{aligned}\right\} \tag{D7}$$

The resulting stress intensity factors, K_{Ia}, K_{IIa}, and K_{IIIa}, as defined above, are represented by K-formulas for the corresponding isotropic boundary value problems, except for cases of non-self-equilibrating loadings. Refer to **Sih (1965b) or Paris (1965)** for complete details.

Finally, for the Hooke's laws defined above, the elastic coefficients, C, relating energy rates to stress intensity factors are

$$\mathcal{G}_i = CK_i^2 \tag{D8}$$

Values of C (for the Case of Plane Stress[1])

Mode i	Isotropic	Orthotropic $A_{16} = A_{26} = A_{45} = 0$	Anisotropic[2]
I	$^1/_E$	$\sqrt{\frac{A_{11}A_{22}}{2}}\left[\sqrt{\frac{A_{22}}{A_{11}}}+\frac{2A_{12}+A_{66}}{2A_{11}}\right]^{1/2}$	$-\frac{A_{11}}{2}\ \mathrm{Im}\left(\frac{\mu_1+\mu_2}{\mu_1\mu_2}\right)$
II	$^1/_E$	$\frac{A_{11}}{\sqrt{2}}\left[\sqrt{\frac{A_{22}}{A_{11}}}+\frac{2A_{12}+A_{66}}{2A_{11}}\right]^{1/2}$	$\frac{A_{11}}{2}\ \mathrm{Im}(\mu_1+\mu_2)$
III	$(1+\nu)/E$	$\frac{1}{2}\sqrt{A_{44}A_{55}}$	$\frac{1}{2}\frac{\left(A_{44}A_{55}-A_{45}^2\right)^{3/2}}{A_{44}A_{55}}$

The resulting values for C have been obtained in the same manner as outlined earlier, Eq. (16) (see page 1.7) but using Eqs. (D1), (D2), and (D3) of this appendix.

[1] **For plane strain (Mode I and Mode II), replace E by $E/\left(1-\nu^2\right)$ for an isotropic material, and replace A_{ij} by $A_{ij} - A_{i3}A_{j3}/A_{33}\,(i,j = 1,2,6)$ for an anisotropic material.**

[2] **For general isotropy, one mode is not decoupled from the other two modes. The expressions given in the table are the coefficients for each mode in the absence of the other two modes.**

APPENDIX E

STRESS INTENSITY FACTORS FOR CRACKS IN A PLATE SUBJECTED TO PINCHING LOADS

Analysis of cracks in an elastic plate near transverse loads (pinching loads) is of practical interest in, for example, accounting for the effect of spot weldings or contriving crack arresters.

The fundamental solution for a pinching force applied near the tip of a crack is presented in **Appendix C**. Namely, the stress intensity factors K_I and K_{II} are given for a semi-infinite crack in a plate subjected to a concentrated pinching force P in plane stress conditions [**Example (4), Appendix C**]. The derivation of this solution is one illustration of the usefulness of the weight function method discussed in **Appendix C**. The solution is collectively given by

$$K_I + i\,K_{II} = \frac{\nu P}{2\sqrt{2\pi}} \cdot \frac{1}{r^{3/2}} e^{-i\frac{3\theta}{2}} \tag{E1}$$

where ν is the Poisson's ratio and the coordinates (r, θ) are shown in Solution (I-1).

This solution serves as the Green's function for a semi-infinite crack subjected to an arbitrarily distributed pinching load. That is, in the (r, θ) coordinate system, if the pinching load distributed on an area S is expressed as $p = p(r, \theta)$, then the stress intensity factors are determined by

$$K_I + i\,K_{II} = \frac{\nu}{2\sqrt{2\pi}} \int_s \frac{p(r,\theta)}{r^{1/2}} e^{-i\frac{3\theta}{2}}\, dr d\theta \tag{E2}$$

Most solutions for a semi-infinite crack in this appendix were obtained by using **Eq. (E2)**.

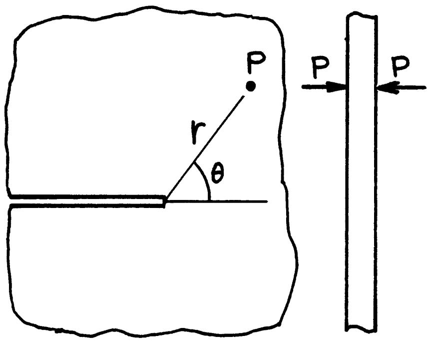
The solutions for concentrated pinching forces for other crack geometries (II-1, II-2, III-1, III-2) were derived using the weight functions in **Appendix C**, and then several additional solutions were obtained by superposition or integration (II-3, II-4; III-3, III-4, etc.).

When the pinching load is a uniform compression over circular areas centered on the line of cracks, the customary method of integrating off the stress distributed in the absence of the cracks seems more convenient than the weight function method. Solutions (I-10), (II-5, 6, 7), and (III-5) were obtained by this method. **Harris (1972)** obtained the complete K_I solution by this method for the entire process for a semi-infinite crack to extend through a circular region under uniform pinching pressure initially located ahead of the crack (Solution I-11). Harris also analyzed the effect of plate thickness (i.e., of the departure from the plane stress conditions).

All solutions included in this appendix are exact in the plane stress conditions and represent the upper bounds of the effect of pinching loads in actual plates.

The following are a few remarks on the crack-arresting effect of pinching loads. Only Mode I is addressed.

1. As observed from Solution (I-1), pinching forces near the crack tip can have the crack-arresting effect $(K_I < 0)$ only when they are applied behind the rays $|\theta| = {}^{\pi}\!/_{3}$ (i.e., in the regions $\frac{\pi}{3} \leq |\theta| \leq \pi$). For example, when a circular area ahead of the crack is uniformly pinched, Solution (I-10), the pinching load contributes positive K_I until the crack extends some distance into the pinched zone. See the sequence of Solutions (I-10), (I-9), and (I-8); and also **Harris (1972)**.
2. The maximum crack-arresting effect is gained when pinching forces are applied on the rays $|\theta| = {}^{2\pi}\!/_{3}$. The dependency on the distance from the crack tip is common for all directions.
3. As readily observed from the comparison of Solutions (I-3) and (I-3a), when the pinching load distribution on a circular area $r \leq R$ is given by a pressure $p(r)$ or a line load $\overline{p}(r)$, for example, Solutions (I-3a), (I-4a), (I-5), and (I-6), the crack-arresting effect becomes 1.5 times as large if the load on the sector $|\theta| \leq {}^{\pi}\!/_{3}$ is removed. Or, if the same total pinching load can be distributed on sectors ${}^{\pi}\!/_{3} \leq |\theta| \leq \pi$ only, the crack-arresting effect becomes 2.25 times as large.

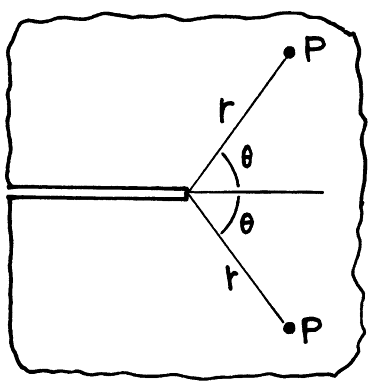


(I-1)

Pinching force P at (r, θ)

$$K_I = \frac{\nu P}{2\sqrt{2\pi}\, r^{3/2}} \cos\frac{3\theta}{2}$$

$$K_{II} = -\frac{\nu P}{2\sqrt{2\pi}\, r^{3/2}} \sin\frac{3\theta}{2}$$

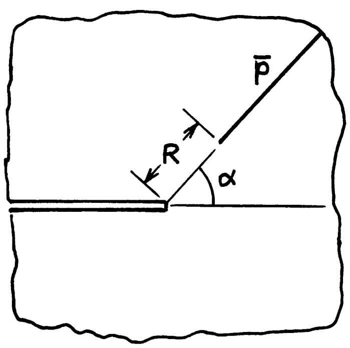


(I-1a)

Pinching forces P at $(r, \pm\theta)$

$$K_I = \frac{\nu P}{\sqrt{2\pi} r^{3/2}} \cos\frac{3\theta}{2}$$

$$K_{II} = 0$$

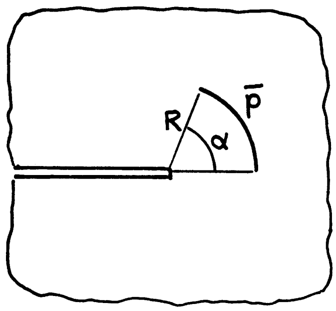


(I-2)

Uniform pinching line force $\overline{p}$ on $r \geq R,\ \theta = \alpha$

$$K_I = \frac{\nu \overline{p}}{\sqrt{2\pi R}} \cos\frac{3\alpha}{2}$$

$$K_{II} = -\frac{\nu \overline{p}}{\sqrt{2\pi R}} \sin\frac{3\alpha}{2}$$

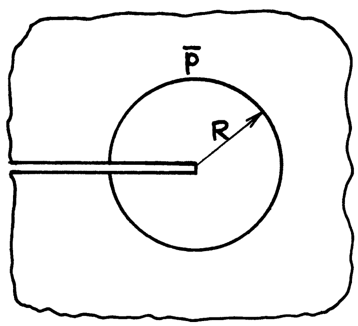


(I-3)

Uniform pinching line force $\overline{p}$ on circular arc $r = R,\ 0 \leq \theta \leq \alpha$

$$K_I = \frac{\nu \overline{p}}{3\sqrt{2\pi R}} \sin\frac{3\alpha}{2}$$

$$K_{II} = -\frac{\nu \overline{p}}{3\sqrt{2\pi R}} \left(1 - \cos\frac{3\alpha}{2}\right)$$

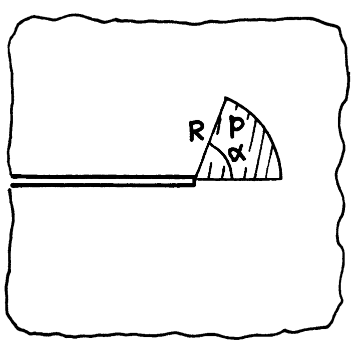


(I-3a)

Uniform pinching line force $\overline{p}$ on circle $r = R$

$$K_I = -\frac{2\nu \overline{p}}{3\sqrt{2\pi R}}$$

$$K_{II} = 0$$

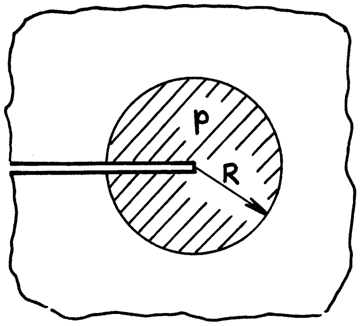


(I-4)

Uniform pinching pressure p on sector $r \leq R,\ 0 \leq \theta \leq \alpha$

$$K_I = \frac{2\nu P}{3\sqrt{2\pi}} \sqrt{R} \sin\frac{3\alpha}{2}$$

$$K_{II} = -\frac{2\nu p}{3\sqrt{2\pi}} \sqrt{R} \left(1 - \cos\frac{3\alpha}{2}\right)$$

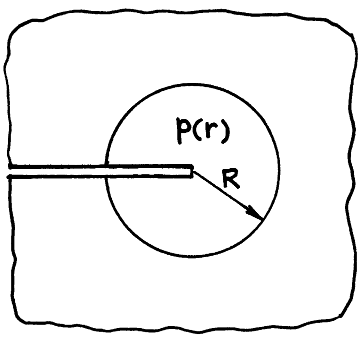


(I-4a)

Uniform pinching pressure p on circular area $r \leq R$

$$K_I = -\frac{4\nu P}{3\sqrt{2\pi}}\sqrt{R}$$

$$K_{II} = 0$$



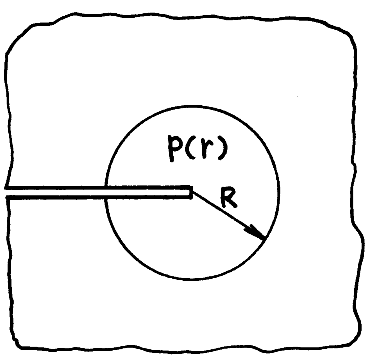
(I-5)

Pinching force P distributed on circular area $r \leq R$

$$p(r) = \frac{P}{2\pi R^2} \cdot \frac{1}{\sqrt{1 - \left({}^{r}\!/_{R}\right)^2}}$$

$$K_I = -\frac{\nu P}{6\sqrt{2\pi} R^{3/2}} \cdot \frac{\Gamma\left(1/4\right)}{\Gamma\left(3/4\right)} = -.1110\frac{\nu P}{R^{3/2}}$$

$$K_{II} = 0$$



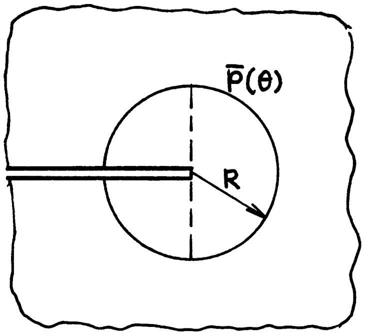
(I-6)

Pinching force P distributed on circular area $r \leq R$

$$p(r) = \frac{3P}{2\pi R^2}\sqrt{1 - \left({}^{r}\!/_{R}\right)^2}$$

$$K_I = -\frac{\nu P}{\sqrt{2\pi} R^{3/2}} \cdot \frac{\Gamma(5/4)}{\Gamma(7/4)} = -\cdot 2220\frac{\nu P}{R^{3/2}}$$

$$K_{II} = 0$$



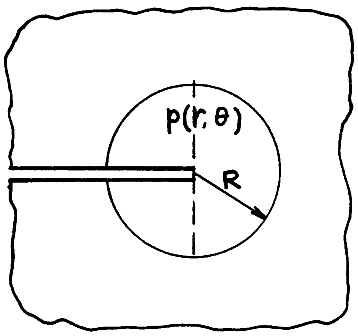
(I-7)

Pinching line force on circle $r = R$

$$\overline{p}(R, \theta) = \overline{p}_0 \cos \theta$$

$$K_I = \frac{6\nu\overline{p}_0}{5\sqrt{2\pi R}}$$

$$K_{II} = 0$$



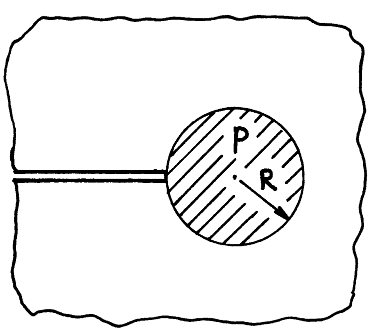
(I-8)

Pinching pressure distributed on circular area $r \leq R$

$$p(r, \theta) = p_0 \frac{r}{R} \cos \theta$$

$$K_I = \frac{4\nu p_0}{5\sqrt{2\pi}} \sqrt{R}$$

$$K_{II} = 0$$

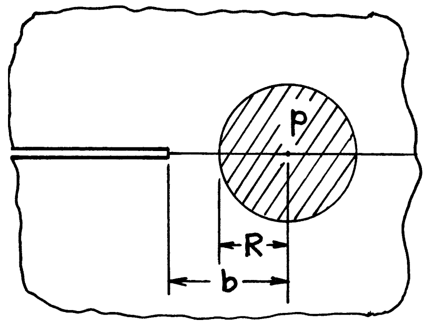


(I-9)

Uniform pinching pressure p on circular area $r \leq 2R \cos \theta$

$$K_I = \frac{\nu p}{2\sqrt{2}} \sqrt{\pi R}$$

$$K_{II} = 0$$

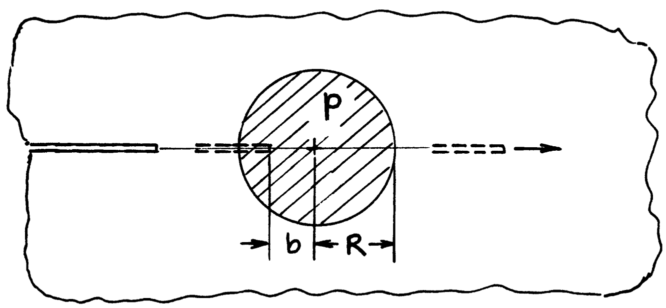


(I-10)

Uniform pinching pressure p on circular area ahead of crack

$$K_I = \frac{\nu p}{2\sqrt{2}}\sqrt{\pi R}\left(\frac{R}{b}\right)^{3/2}$$

$$K_{II} = 0$$



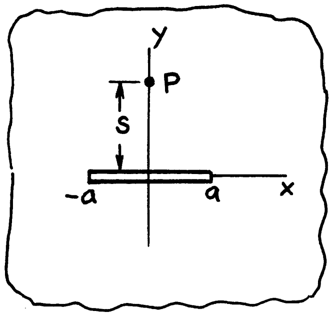
(I-11)

Crack in (I-10) extending through circular zone under uniform pinching pressure p

$$K_I = \nu p\sqrt{R}\, f\left(\frac{b}{R}\right)$$

$$K_{II} = 0$$

Complete solution $f(^b/_R)$ is given in **Harris (1972)**. (I-4a), (I-9), and (I-10) are special cases. Effect of plate thickness is also accounted for in **Harris (1972)**.

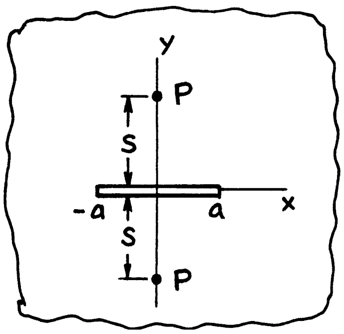


(II-1)

Pinching force P at $(0, s)$

$$K_I = -\frac{\nu P}{2}\sqrt{\frac{a}{\pi}}\frac{s}{\left(a^2 + s^2\right)^{3/2}}$$

$$K_{II\pm a} = \pm\frac{\nu P}{2}\sqrt{\frac{a}{\pi}}\frac{a}{\left(a^2 + s^2\right)^{3/2}}$$

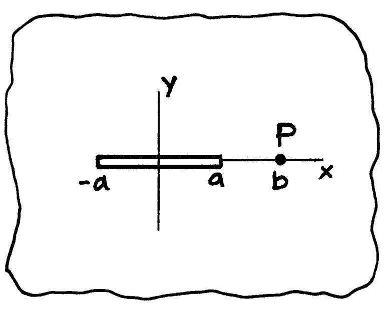


(II-1a)

Pinching forces P at $(0, \pm s)$

$$K_I = -\nu P\sqrt{\frac{a}{\pi}}\frac{s}{\left(a^2 + s^2\right)^{3/2}}$$

$$K_{II} = 0$$

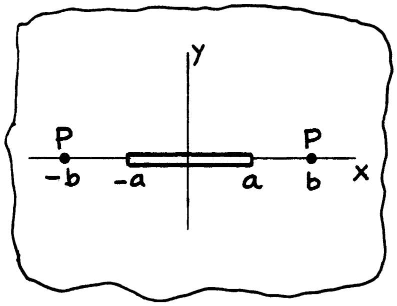


(II-2)

Pinching force P at $(b, 0)$

$$K_{I\pm a} = \pm\frac{\nu P}{2}\sqrt{\frac{a}{\pi}}\frac{a}{(b \mp a)\sqrt{b^2 - a^2}}$$

$$K_{II} = 0$$

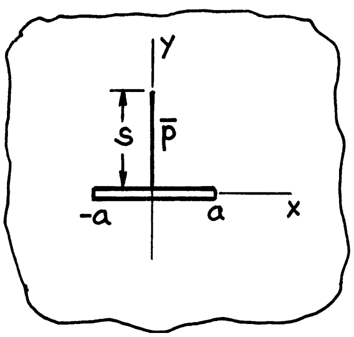


(II-2a)

Pinching forces P at $(\pm b, 0)$

$$K_I = \nu P \sqrt{\frac{a}{\pi}} \frac{b}{\left(b^2 - a^2\right)^{3/2}}$$

$$K_{II} = 0$$

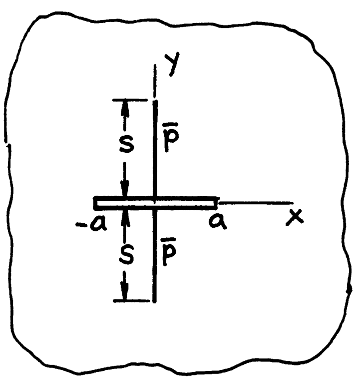


(II-3)

Uniform pinching line force $\bar{p}$ on $x = 0$, $0 \leq y \leq s$

$$K_I = -\frac{\nu \bar{p}}{2\sqrt{\pi a}} \left(1 - \frac{a}{\sqrt{a^2 + s^2}}\right)$$

$$K_{II \pm a} = \pm \frac{\nu \bar{p}}{2\sqrt{\pi a}} \frac{s}{\sqrt{a^2 + s^2}}$$

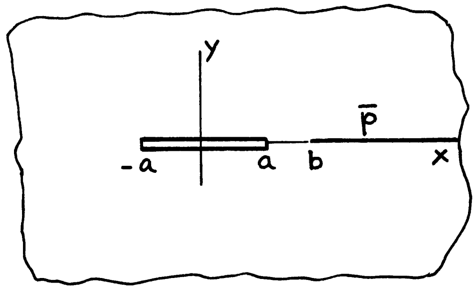


(II-3a)

Uniform pinching line force $\bar{p}$ on $x = 0$, $|y| \leq s$

$$K_I = -\frac{\nu \bar{p}}{\sqrt{\pi a}} \left(1 - \frac{a}{\sqrt{a^2 + s^2}}\right)$$

$$K_{II} = 0$$

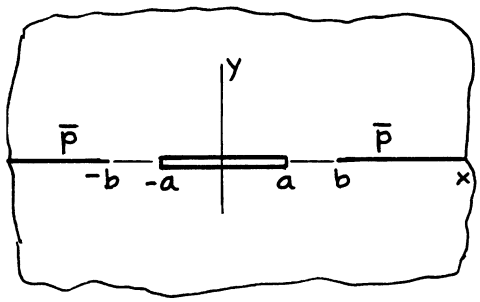


(II-4)

Uniform pinching line force $\bar{p}$ on $x \geq b$, $y = 0$

$$K_{I_{\pm a}} = \pm \frac{\nu \bar{p}}{2\sqrt{\pi a}} \left(\sqrt{\frac{b \pm a}{b \mp a}} - 1 \right)$$

$$K_{II} = 0$$

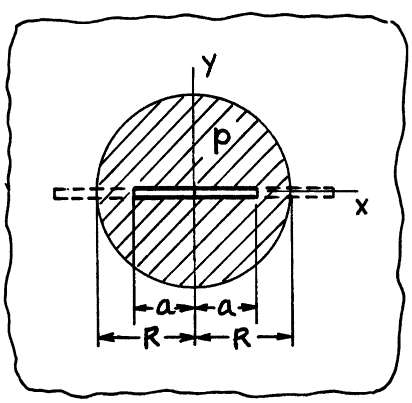


(II-4a)

Uniform pinching line force $\bar{p}$ on $|x| \geq b$, $y = 0$

$$K_I = \frac{\nu \bar{p}}{\sqrt{\pi a}} \frac{a}{\sqrt{b^2 - a^2}}$$

$$K_{II} = 0$$



(II-5)

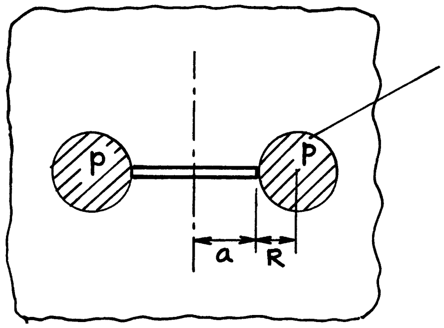
Uniform pinching pressure p on circle $r \leq R$

$$a \leq R: \; K_I = -\frac{\nu P}{2}\sqrt{\pi a}$$

$$K_{II} = 0$$

$$a \geq R: \; K_I = -\frac{\nu P}{2}\sqrt{\pi a} \cdot \frac{2}{\pi}\left[\sin^{-1}\frac{R}{a} - \frac{R}{a}\sqrt{1 - \left(\frac{R}{a}\right)^2} \right]$$

$$K_{II} = 0$$

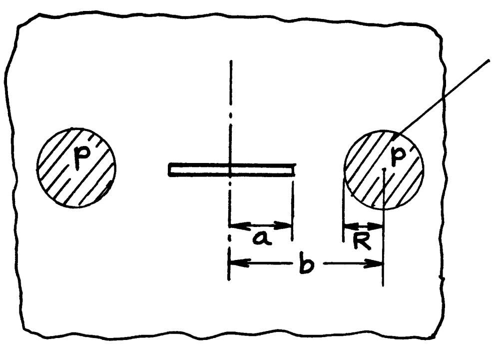


Uniform pinching pressure p

$$K_I = \nu p\sqrt{\pi a}\ \frac{\left(\frac{R}{a+R}\right)^2}{\left[1-\left(\frac{a}{a+R}\right)^2\right]^{3/2}}$$

$$K_{II} = 0$$

(II-6)



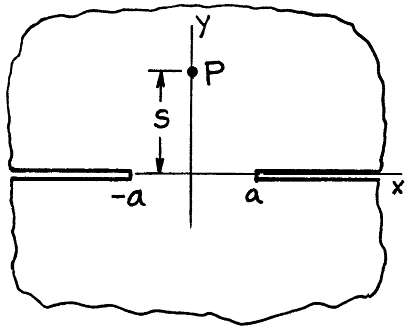
Uniform pinching pressure p

$$K_I = \nu p\sqrt{\pi a}\ \frac{bR^2}{\left(b^2-a^2\right)^{3/2}}$$

$$K_{II} = 0$$

(II-6) is a special case with $b = a + R$.

(II-7)

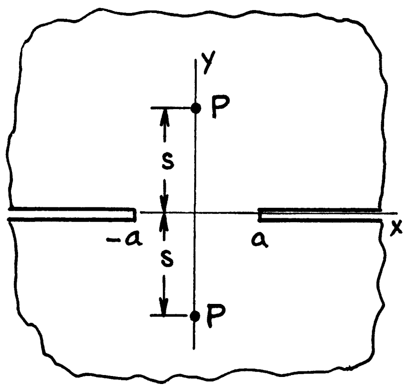


Pinching force P at $(0, s)$

$$K_I = \frac{\nu P}{2}\sqrt{\frac{a}{\pi}}\frac{a}{\left(a^2 + s^2\right)^{3/2}}$$

$$K_{II} = \frac{\nu P}{2}\sqrt{\frac{a}{\pi}}\frac{s}{\left(a^2 + s^2\right)^{3/2}}$$

(III-1)

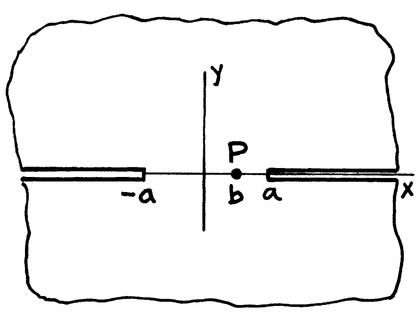


Pinching forces P at $(0, \pm s)$

$$K_I = \nu P\sqrt{\frac{a}{\pi}}\frac{a}{\left(a^2 + s^2\right)^{3/2}}$$

$$K_{II} = 0$$

(III-1a)

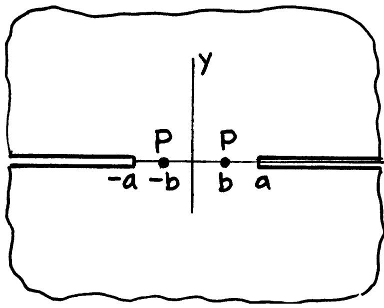


Pinching force P at $(b, 0)$

$$K_{I\pm a} = \frac{\nu P}{2}\sqrt{\frac{a}{\pi}}\frac{1}{(a \mp b)\sqrt{a^2 - b^2}}$$

$$K_{II} = 0$$

(III-2)

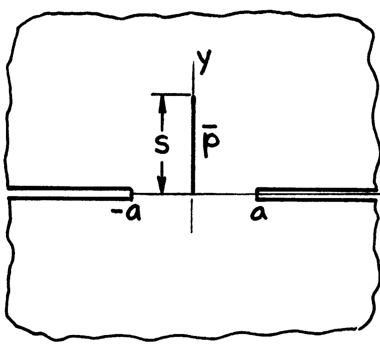


(III-2a)

Pinching forces P at $(\pm b, 0)$

$$K_I = \nu P\sqrt{\frac{a}{\pi}}\frac{a}{\left(a^2 - b^2\right)^{3/2}}$$

$$K_{II} = 0$$

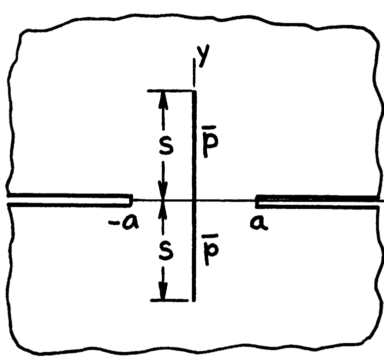


(III-3)

Uniform pinching line force $\overline{p}$ on $x = 0$, $0 \leq y \leq s$

$$K_I = \frac{\nu \overline{p}}{2\sqrt{\pi a}}\frac{s}{\sqrt{a^2 + s^2}}$$

$$K_{II\pm a} = \pm\frac{\nu \overline{p}}{2\sqrt{\pi a}}\left(1 - \frac{a}{\sqrt{a^2 + s^2}}\right)$$

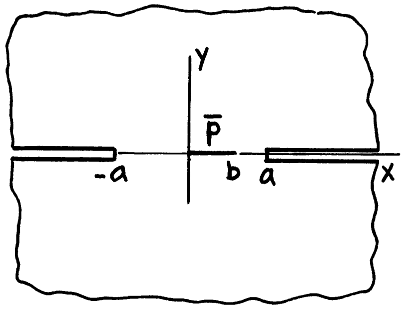


(III-3a)

Uniform pinching line force $\overline{p}$ on $x = 0$, $|y| \leq s$

$$K_I = \frac{\nu \overline{p}}{\sqrt{\pi a}}\frac{s}{\sqrt{a^2 + s^2}}$$

$$K_{II} = 0$$

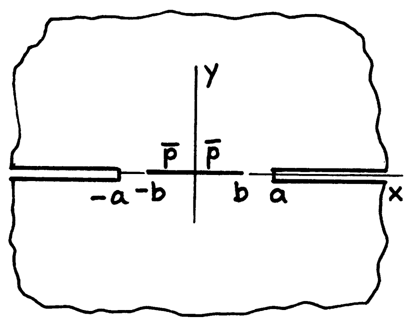


(III-4)

Uniform pinching line force $\overline{p}$ on $0 \leq x \leq b, y = 0$

$$K_{I\pm a} = \pm \frac{\nu \overline{p}}{2\sqrt{\pi a}}\left(\sqrt{\frac{a \pm b}{a \mp b}} - 1\right)$$

$$K_{II} = 0$$

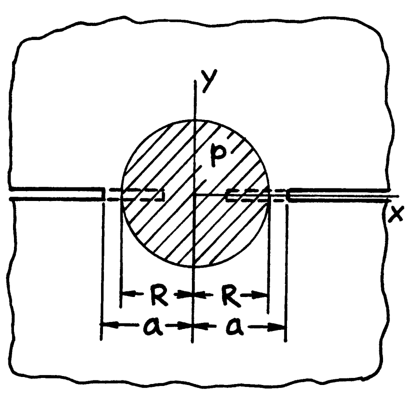


(III-4a)

Uniform pinching line force $\overline{p}$ on $|x| \leq b, y = 0$

$$K_I = \frac{\nu \overline{p}}{\sqrt{\pi a}} \frac{b}{\sqrt{a^2 - b^2}}$$

$$K_{II} = 0$$



(III-5)

Uniform pinching pressure p on circular area $r \leq R$

$$a \geq R : K_I = \frac{\nu P}{2}\sqrt{\pi a}\left(\frac{R}{a}\right)^2$$

$$K_{II} = 0$$

$$a \leq R : \; K_I = \frac{\nu P}{2}\sqrt{\pi a}\left(\frac{R}{a}\right)^2 \cdot \frac{2}{\pi}\left[\sin^{-1}\frac{a}{R} - \frac{a}{R}\sqrt{1 - \left(\frac{a}{R}\right)^2}\right]$$

$$K_{II} = 0$$

NOTE: K_I is always positive.

APPENDIX F

CRACKS IN RESIDUAL STRESS FIELDS

Analyses of cracks developed in residual stress fields (including thermal stress fields, etc.) are of practical importance in welded structures and in many other applications. The residual stress field is, in the absence of external loads, a self-balanced, built-in field. Determination of the intensity of the crack-tip elastic field for cracks introduced into the residual stress field requires no special treatment. The method is based on the superposition principle.

The following two stress states should be superimposed to satisfy the free surface condition on the crack:

a. The residual stress field in the absence of the crack
b. The opening pressure applied on the crack surfaces whose distribution corresponds to a.

That is, the effect of the presence of the crack on the resultant stress field is determined solely from stress state b. For example, the determination of stress intensity factor K_I is reduced to no more than the evaluation of an integral if the Green's function for K_I (K_I solution for a pair of concentrated splitting forces on crack surfaces) is known for the crack geometry of interest. For the examples on **pages F.2 through F.5** in this appendix, the K_I solution on **page 5.11** is most conveniently used as the Green's function K_{IG}.

$$K_{IG}(a,\, b;\, P) = \frac{2P}{\sqrt{\pi a}} \cdot \frac{1}{\sqrt{1 - \left(b/a\right)^2}} \qquad (\mathbf{F1})$$

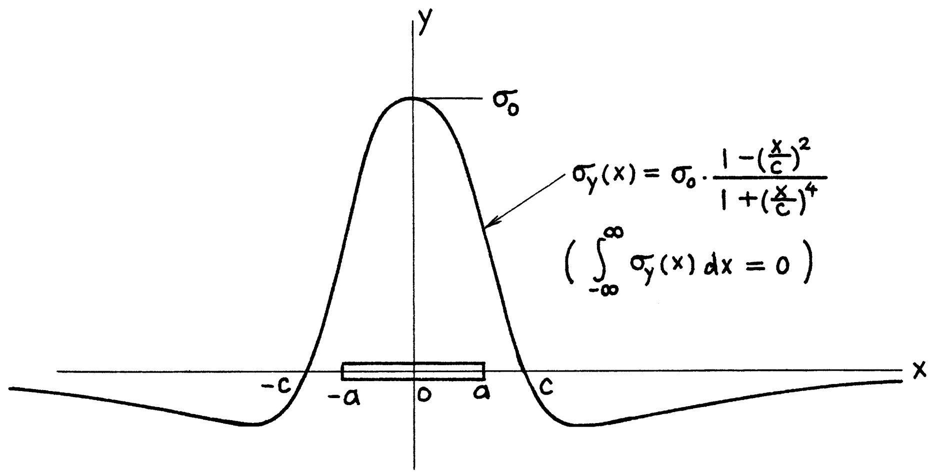
Then the stress intensity factor for the crack absent stress distribution $\sigma_y(x)$ is readily determined by integration.

$$K_I = \int_0^a K_{IG}(a,\, x;\, \sigma_y(x)dx)$$

$$= \frac{2}{\sqrt{\pi a}} \int_0^a \frac{\sigma_y(x)\, dx}{\sqrt{1 - (x/a)^2}}$$

Similarly, for displacement solutions (crack opening area, crack opening displacement, etc.), the solutions on **page 5.11a** may be used as Green's functions. Use of Paris' equation (see Appendix B), however, is often more convenient (i.e., integrations are simpler) for this purpose.

The following pages contain solutions for various "crack-absent" residual stress distributions and crack geometries. An actual, practical situation may be approximately represented by one of these solutions. When additional external loads are also present, the total solution is obtained by superposition.



$$\alpha = \frac{a}{c}$$

$$K_I = \sigma_0\sqrt{\pi a}\left\{\frac{\sqrt{1+\alpha^4}-\alpha^2}{1+\alpha^4}\right\}^{1/2}$$

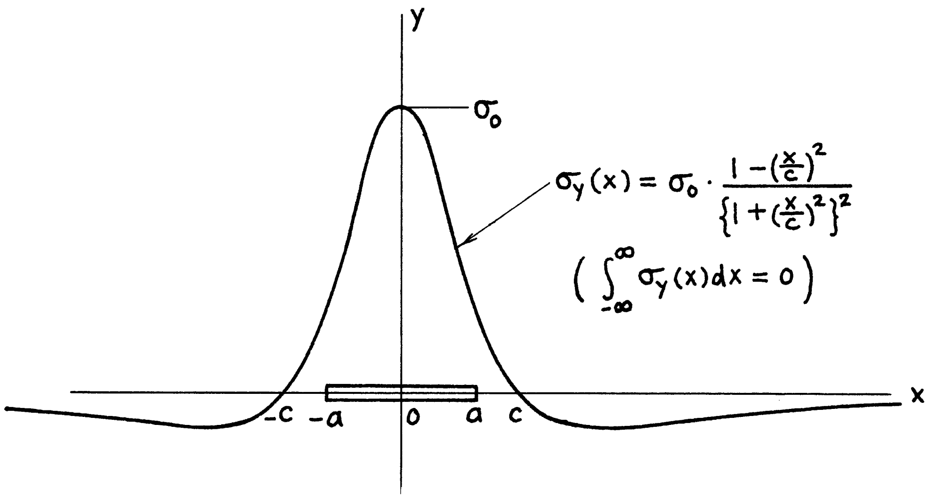
Crack Opening Area:

$$A = \frac{4\sigma_0 \pi c^2}{E'}\left\{1-\left(\sqrt{1+\alpha^4}-\alpha^2\right)^{1/2}\right\}$$

Opening at Center: $\delta = 2v(0,0)$

$$\delta = \frac{4\sigma_0\, c}{E'}\cdot\frac{1}{\sqrt{2}}\cos^{-1}\left(\sqrt{1+\alpha^4}-\alpha^2\right)$$

Method: K_I Integration of **page 5.11**
A and δ Paris' Equation (see **Appendix B**)
Accuracy: Exact
References: **Tada 1983b, 1985**



$$\alpha = \frac{a}{c}$$

$$K_I = \sigma_0 \sqrt{\pi a} \frac{1}{\left(1 + \alpha^2\right)^{3/2}}$$

Crack Opening Area:

$$A = \frac{4\sigma_0 \pi c^2}{E'} \left\{ 1 - \frac{1}{\sqrt{1 + \alpha^2}} \right\}$$

Opening at Center: $\delta = 2v(0, 0)$

$$\delta = \frac{4\sigma_0 a}{E'} \cdot \frac{1}{\sqrt{1 + \alpha^2}}$$

Method: K_I Integration of **page 5.11**
A and δ Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**